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MIL-M-28787/363
23 July 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

CONTROLLER/TIMING MODULE, KEY CODE CEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic controller/timing module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 1.0700 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CEM.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 160 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cord	Parameter 5/	Ref test table	Test type 1/-3/	25 + 5°C 4/		0 + 5°C 4/		60 + 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation									
	initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	S	1	INF	1	INF	1	INF	MOhm
A2-A34	Isolation									
	initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B5-BP	Continuity	III	Q	0.0	0.1	0.0	0 1	0.0	0.1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests-------						
D1-D26	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D27-D30	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D31	Ibd	V	S	0.0	2.0	0.0	2 0	0.0	2.0	mA
D32	Ibd	V	S	0.0	0.4	0.0	0 4	0.0	0.4	mA
D33, D34	Ibd	V	S	0.0	0.111	0 0	0 111	0 0	0 111	mA
E1-E34	Vio	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
F1-F8	Ios	VII	S	-100	-15	-100	-15	-100	-15	mA
F9-F.6	Ios	VII	S	-225	-40	-225	-40	-225	-40	mA
G1-G26	Iih	VIII	X	0	20	0	20	0	20	uA
G27, G28	Iih	VIII	X	-294	-247	-294	-247	-294	-247	uA
G29	Iih	VIII	X	-150	100	-150	100	-150	100	uA
G30-G33	Iih	VIII	X	0	25	0	25	0	25	uA
G34	Iih	VIII	X	0	50	0	50	0	50	uA
H1-H20	Iil	IX	X	-200	0	-200	0	-200	0	uA
H21-H26	Iil	IX	X	-400	-30	-400	-30	-400	-30	uA
H27-H30	Iil	IX	X	-250	0	-250	0	-250	0	uA
H31	Iil	IX	X	-500	0	-500	0	-500	0	uA
H32, H33	Iil	IX	X	-742	-480	-742	-480	-742	-480	uA
H34	Iil	IX	X	-2400	-350	-2400	-350	-2400	-350	uA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 + 5°C 4/		0 + 5°C 4/		60 + 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
J1-J25	Voh	X	X	2.4	4.5	2 4	4.5	2.4	4.5	V
J26-J33	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J34, J35	Voh	X	X	2.4	2.6	2.4	2.6	2.4	2.6	V
K1-K16	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K17-K33	Vol	XI	X	0.0	0 5	0 0	0.5	0.0	0.5	V
L1-L8	Iozh	XII	X	-20	20	-20	20	-20	20	uA
L9-L19	Iozh	XII	X	-10	10	-10	10	-10	10	uA
M1-M8	Iozl	XIII	X	-200	20	-200	20	-200	20	uA
M9-M19	Iozl	XIII	X	-10	10	-10	10	-10	10	uA
N	Icc	XIV	X	70	696	70	696	70	696	mA
P1, P2	Functional	XV	X	See table XV						
Q1, Q2	Dynamic	XVI	S	See table XVI						
Q3, Q4	Dynamic	XVI	Q	See table XVI						
R	PSFC	XVIII	S	4.2	7.0	4.2	7.0	4.2	7.0	uF
S	PSTA	XIX	Q	0	400	0	400	0	400	mV
T1-T20	Output crosstalk, high level	XX	Q	0	400	0	400	0	400	mV
U1-U20	Output crosstalk, low level	XXI	Q	0	400	0	400	0	400	mV

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
V1-[illegible]	Cin	XXII	Q	2	25	2	25	2	25	pF
V8-[illegible]16	Cin	XXII	Q	2	15	2	15	2	15	pF
V17-[illegible]24	Cin	XXII	Q	2	25	2	25	2	25	pF
V25	Cin	XXII	Q	2	50	2	50	2	50	pF
V26	Cin	XXII	Q	4	70	4	70	4	70	pF
W1-[illegible]3	Cout	XXIII	Q	2	25	2	25	2	25	pF
W4,[illegible]5	Cout	XXIII	Q	2	40	2	40	2	40	pF
W6-[illegible]13	Cout	XXIII	Q	2	20	2	20	2	20	pF
W14-[illegible]29	Cout	XXIII	Q	2	30	2	30	2	30	pF
W30-[illegible]31	Cout	XXIII	Q	2	60	2	60	2	60	pF
W32	Cout	XXIII	Q	2	50	2	50	2	50	pF
W33	Cout	XXIII	Q	4	85	4	85	4	85	pF
X	Threshold	XXIV	X	See table XXIV						
Y1-[illegible]8	Disable time	XXV	Q	0	42	0	42	0	42	nS
Y9-[illegible]24	Disable time	XXV	Q	0	75	0	75	0	75	nS

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/363. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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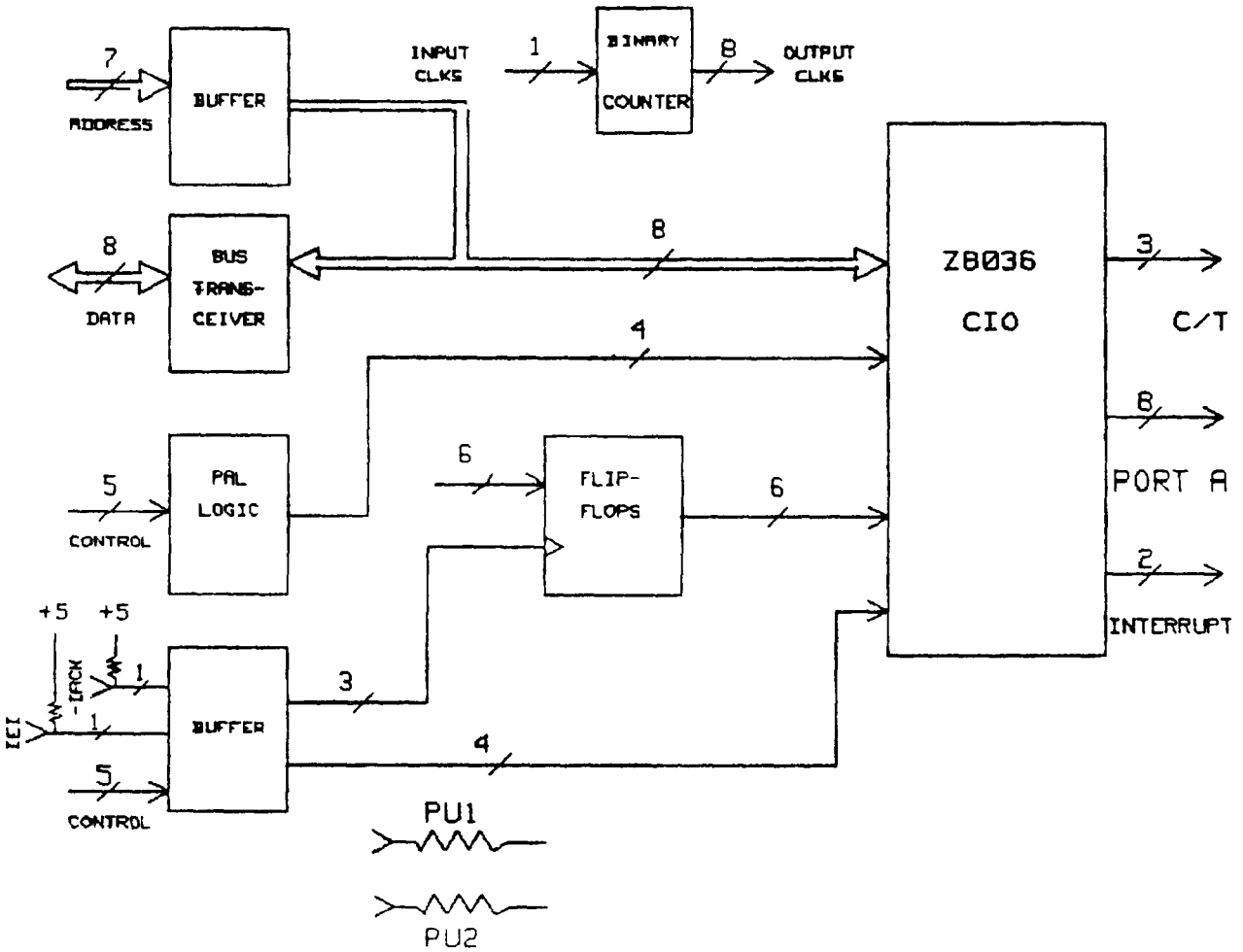


FIGURE 1a. CEM functional block diagram.

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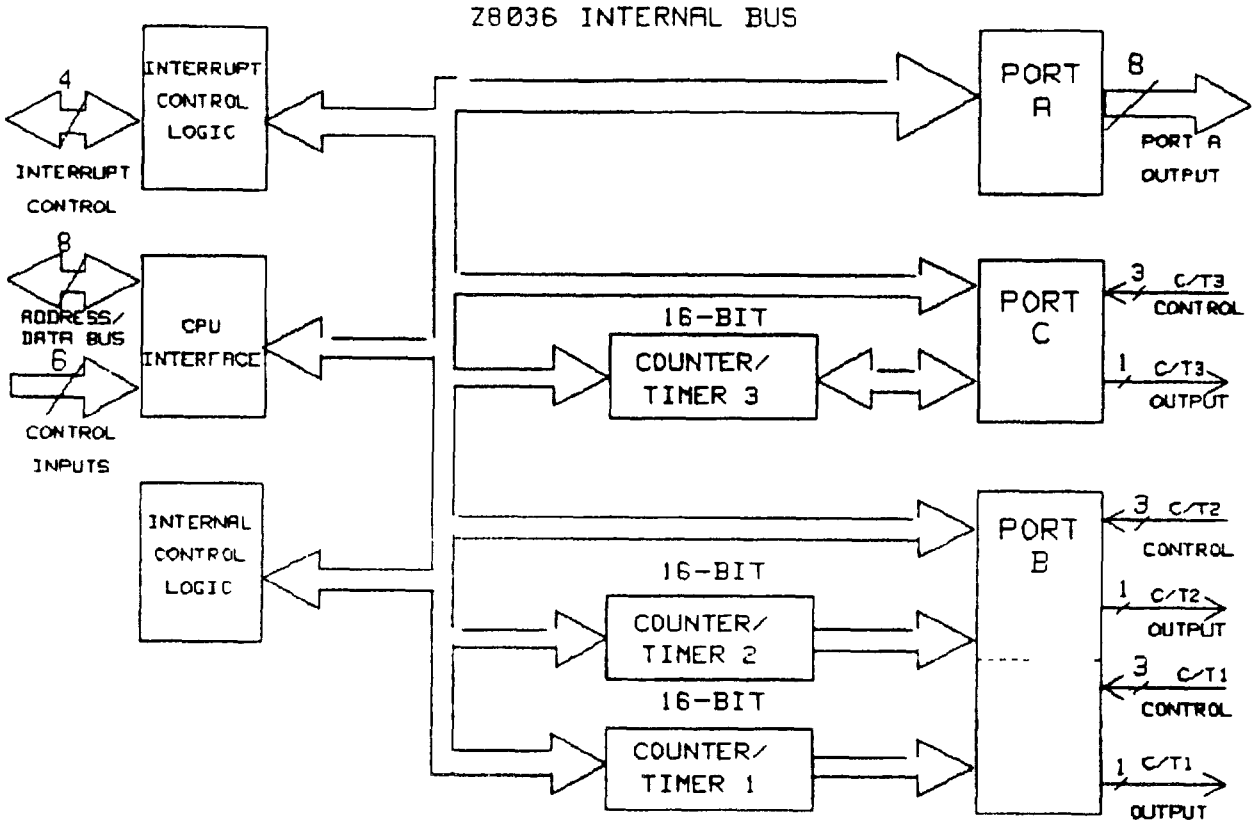


FIGURE 1b. CEM functional block diagram (CIO block diagram).

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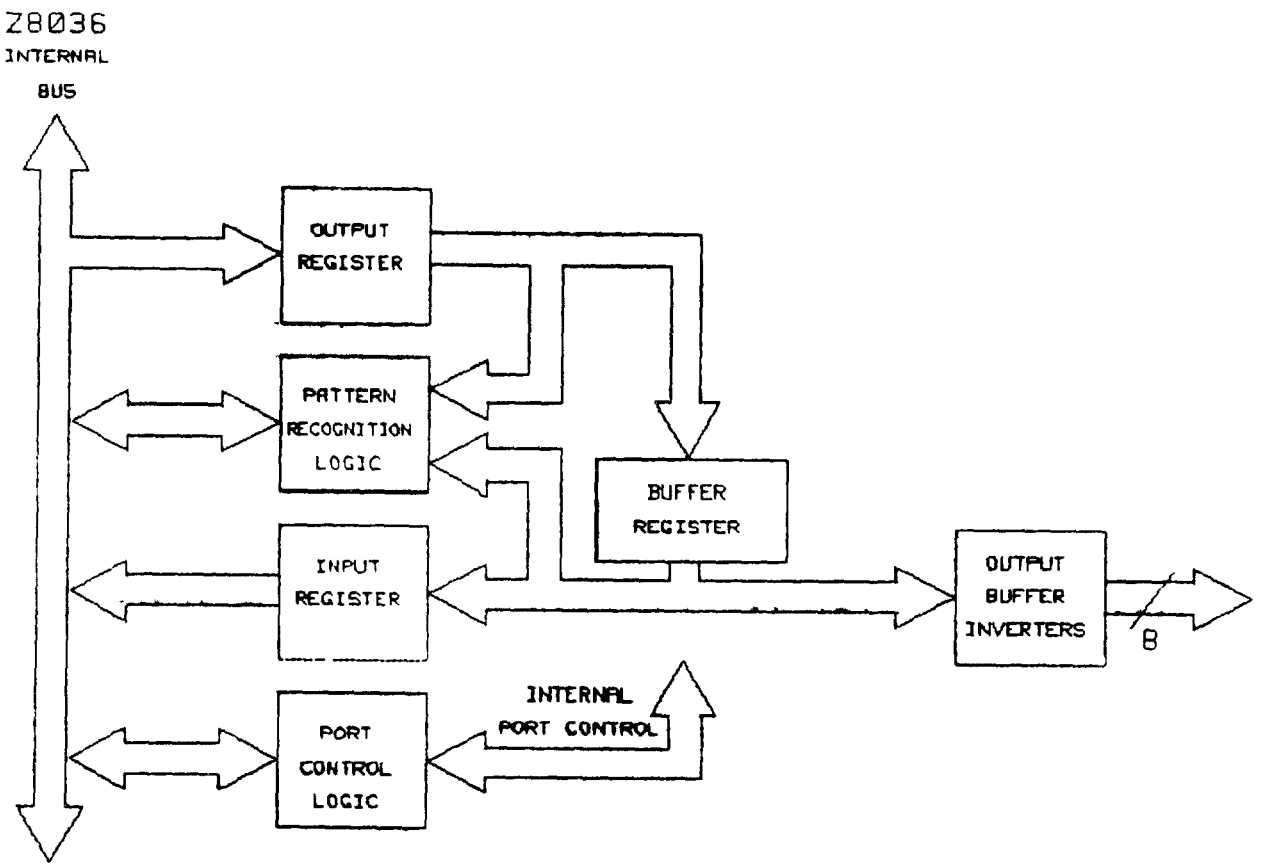


FIGURE 1c. CEM functional block diagram (port A).

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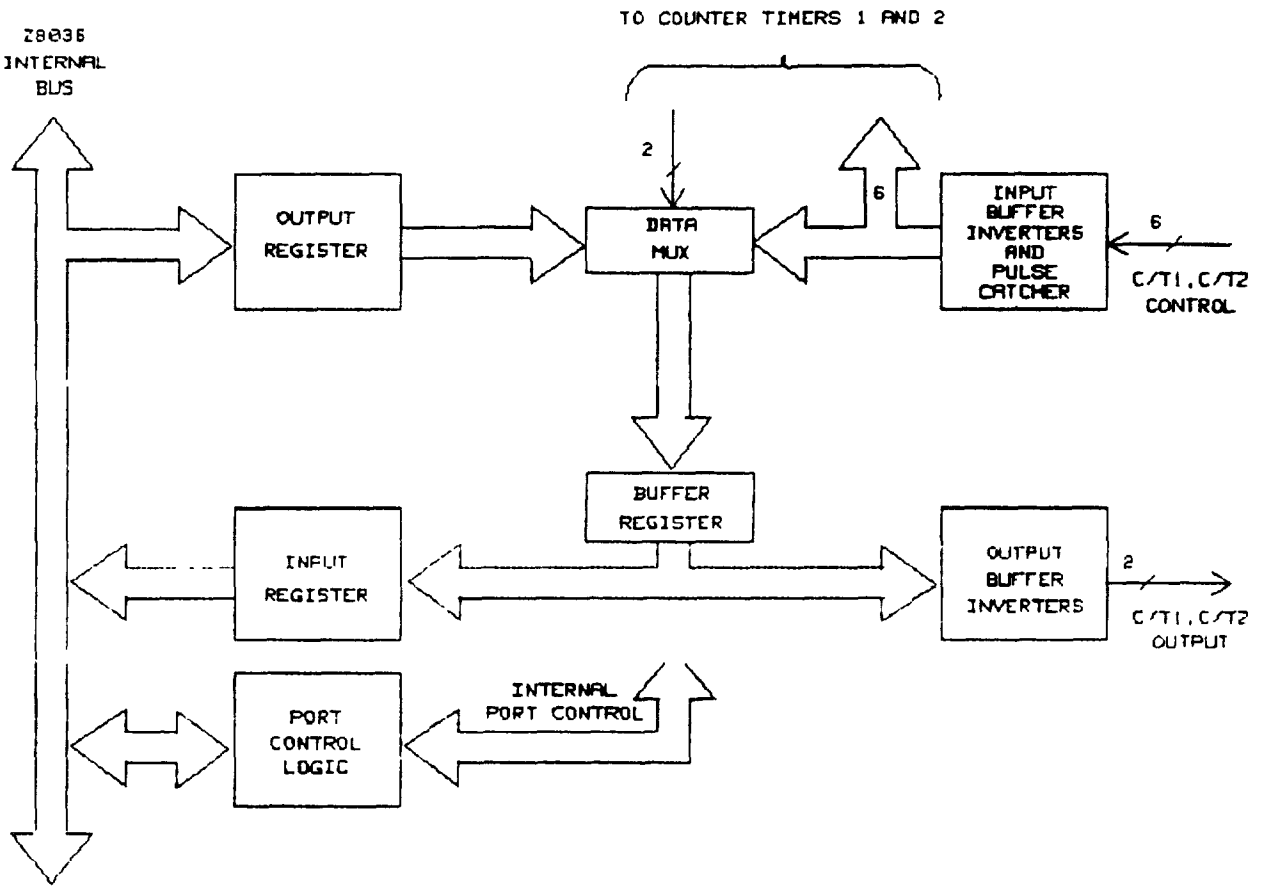


FIGURE 1d. CEM functional block diagram (port B).

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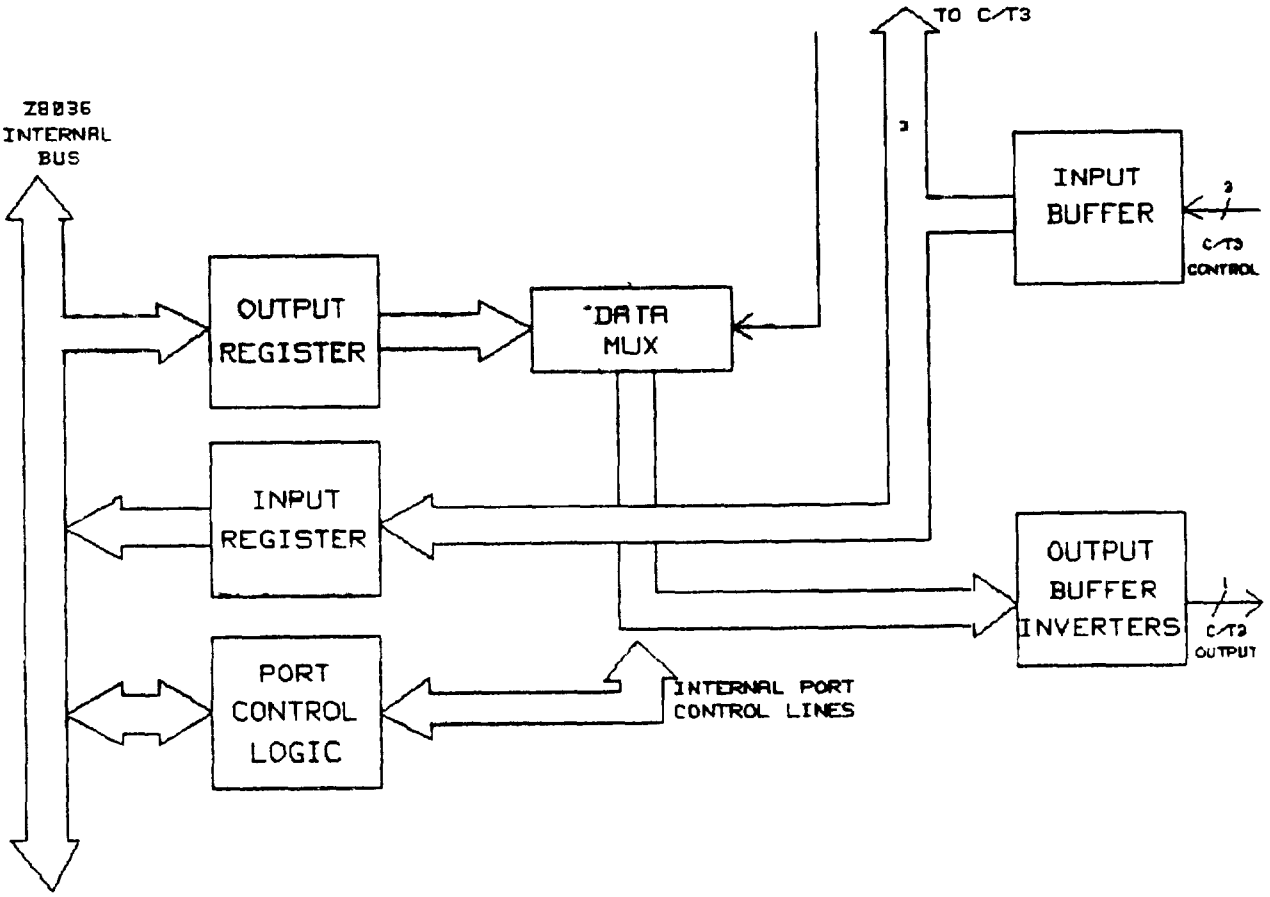


FIGURE 1e. CEM functional block diagram (port C).

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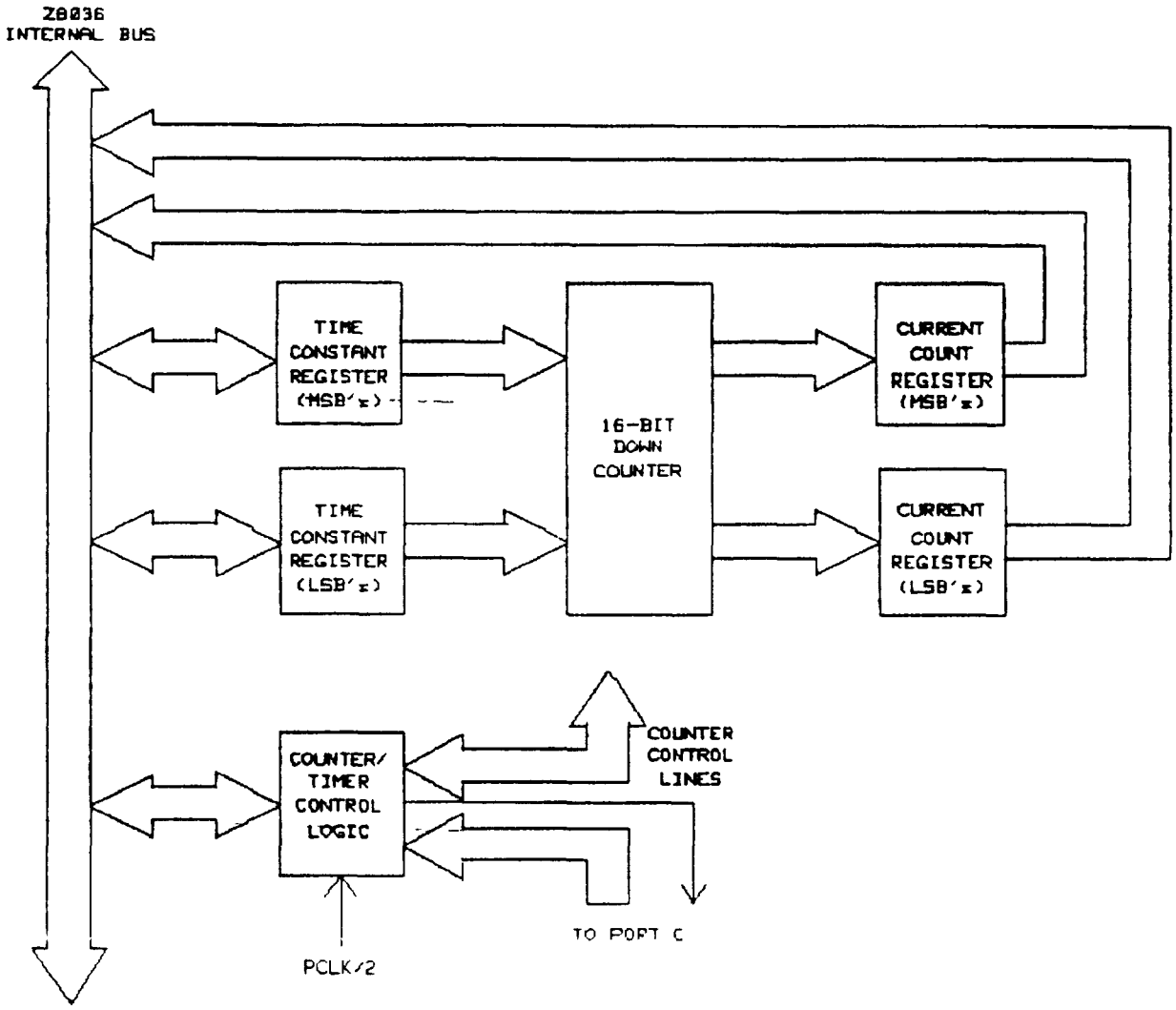


FIGURE 1f. CEM functional block diagram (counter/timer).

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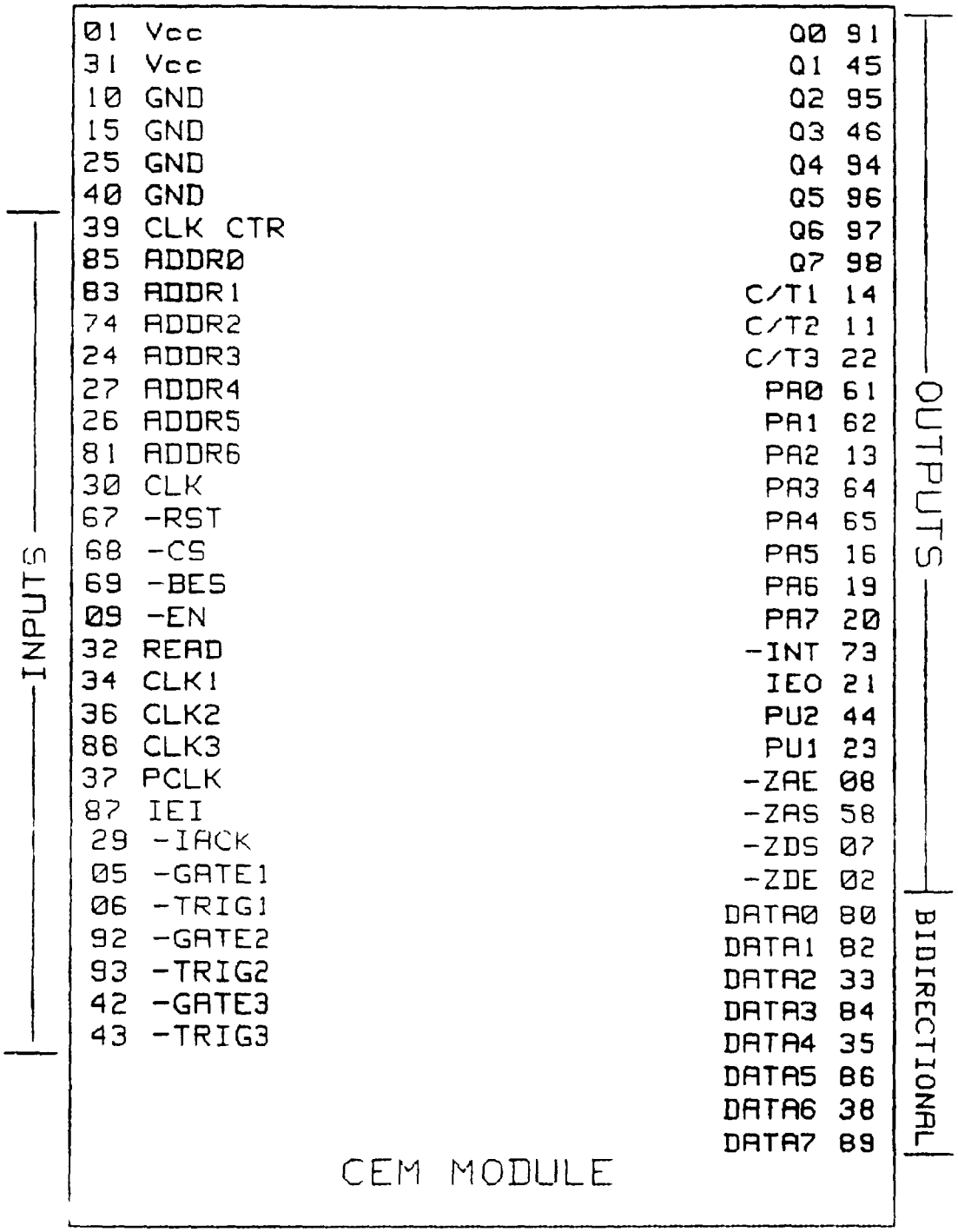


FIGURE 2. CEM pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

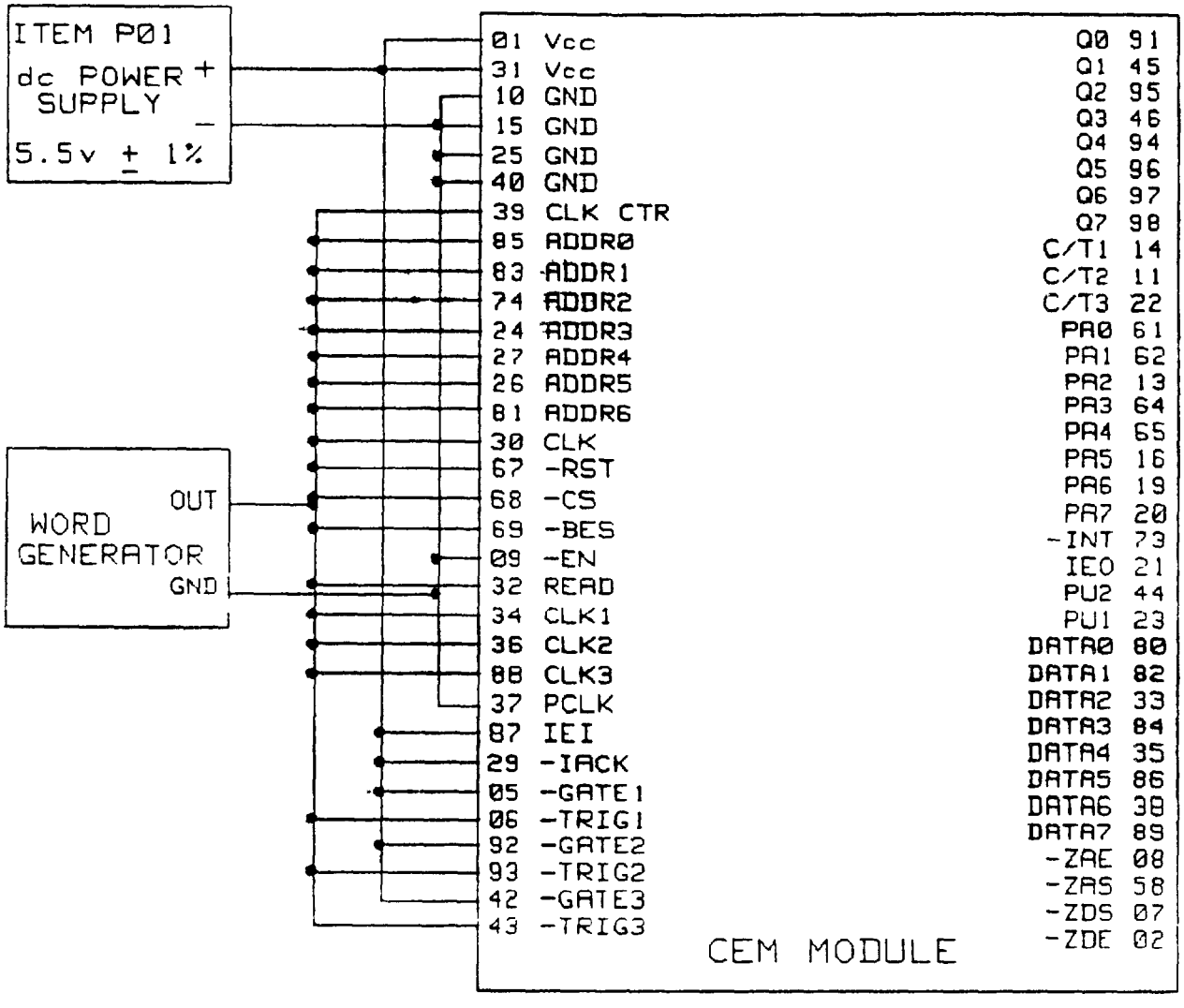
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through U and X shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 4 and 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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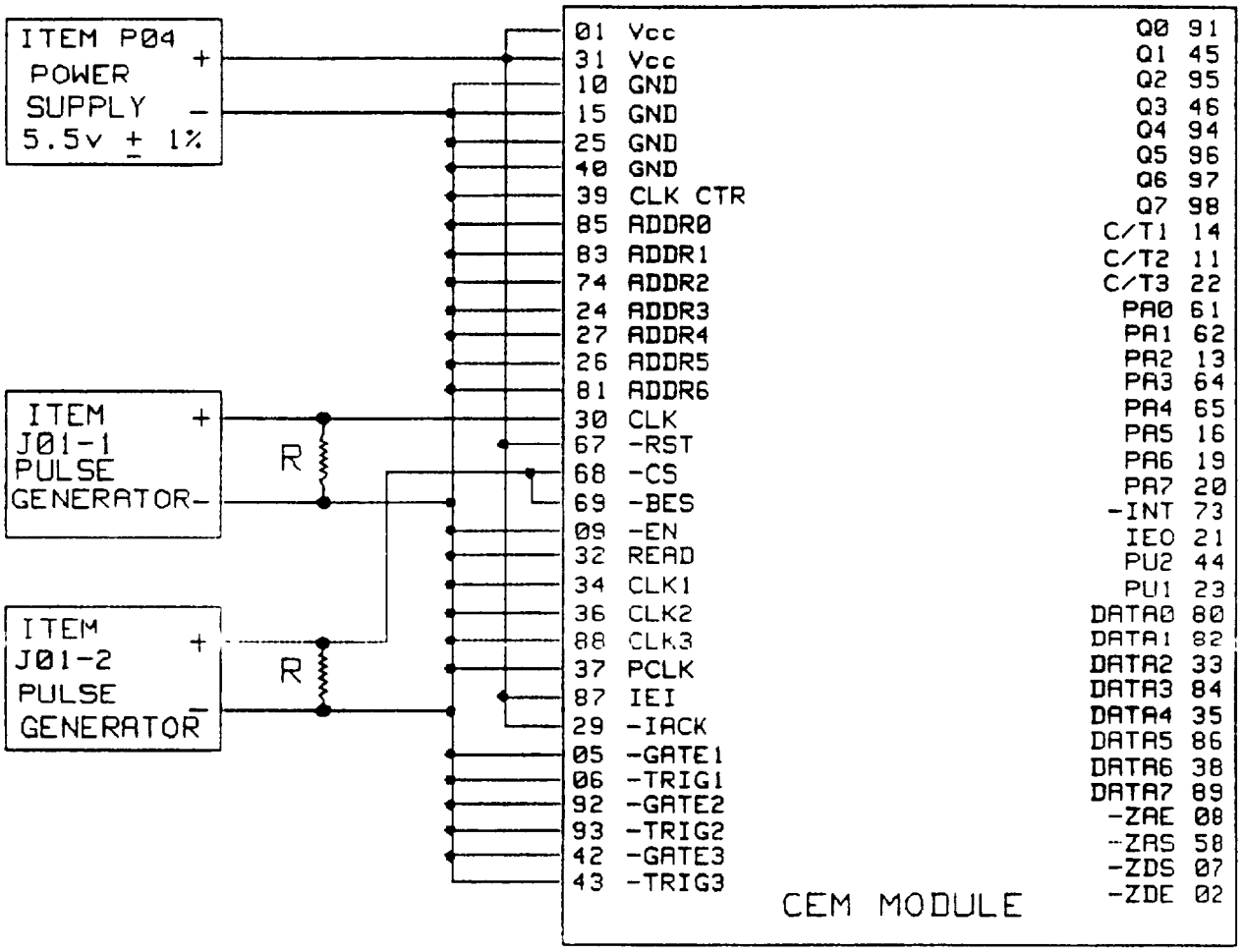


NOTES:

1. Loads are defined in the output voltage tests.
2. Word generator must also be applied to DATA0-DATA7

FIGURE 3. CEM operating temperature and life test circuit.

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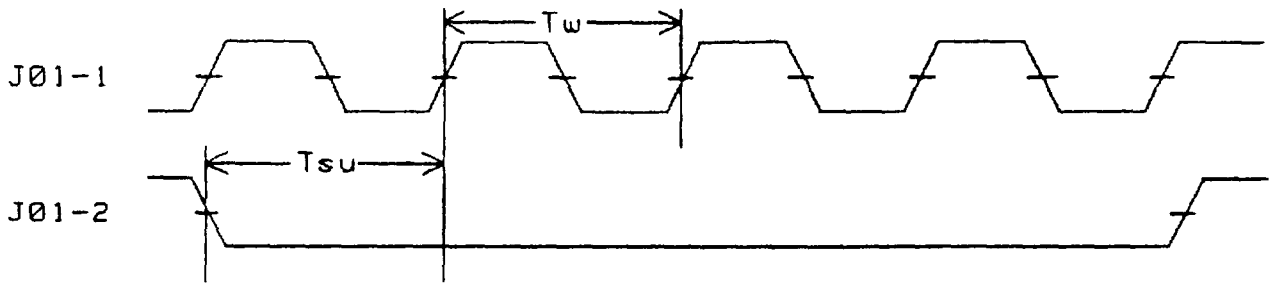
NOTES:

1. Loads are defined in the output voltage tests.

2. R = 51 Ohms ± 5 percent.

FIGURE 4. CEM component temperature test circuit.

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NOTES:

1. Input pulse parameters for J01-1 and J01-2:

   Tw = 200 nS minimum.
   Tsu = 30 nS minimum.

FIGURE 5. Component temperature test waveforms.

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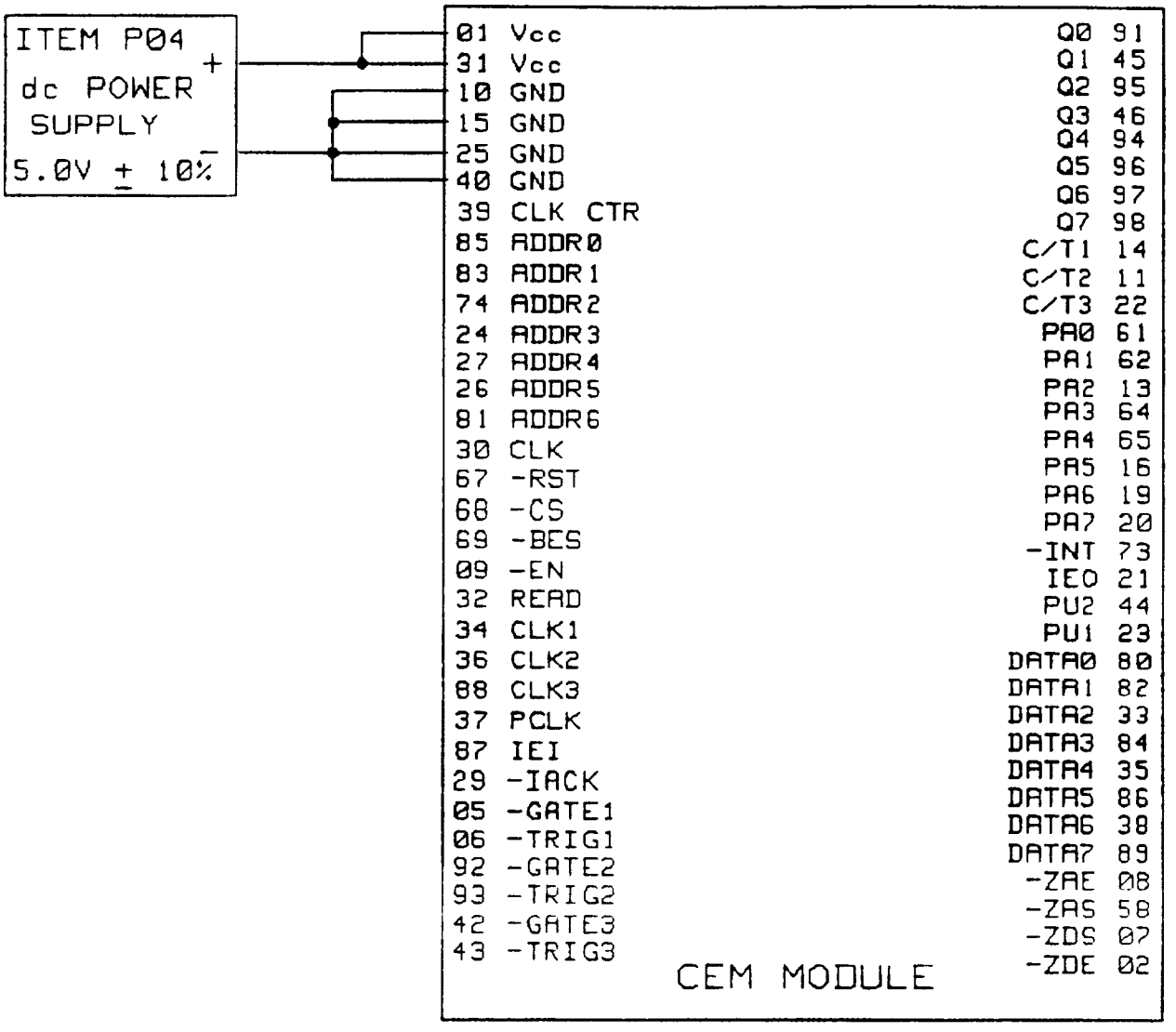


FIGURE 6. CEM vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B, G through U and X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through U and X shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioff = open drain leakage current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.83 W (maximum), 2.27 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
ADDR0	-1.5	0.7	2.0	-0.2	20	100	25
ADDR1	-1.5	0.7	2.0	-0.2	20	100	25
ADDR2	-1.5	0.7	2.0	-0.2	20	100	25
ADDR3	-1.5	0.7	2.0	-0.2	20	100	25
ADDR4	-1.5	0.7	2.0	-0.2	20	100	25
ADDR5	-1.5	0.7	2.0	-0.2	20	100	25
ADDR6	-1.5	0.7	2.0	-0.2	20	100	25
READ	-1.5	0.7	2.0	-0.2	20	100	25
CLK1	-1.5	0.7	2.0	-0.2	20	100	25
CLK2	-1.5	0.7	2.0	-0.2	20	100	25
CLK3	-1.5	0.7	2.0	-0.2	20	100	25
PCLK	-1.5	0.7	2.0	-0.2	20	100	25
DATA0	-1.5	0.7	2.0	-0.2	20	100	30

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DATA1	-1.5	0.7	2.0	-0.2	20	100	30
DATA2	-1.5	0.7	2.0	-0.2	20	100	30
DATA3	-1.5	0.7	2.0	-0.2	20	100	30
DATA4	-1.5	0.7	2.0	-0.2	20	100	30
DATA5	-1.5	0.7	2.0	-0.2	20	100	30
DATA6	-1.5	0.7	2.0	-0.2	20	100	30
DATA7	-1.5	0.7	2.0	-0.2	20	100	30
-GATE1	-1.5	0.7	2.0	-0.4	20	100	15
-GATE2	-1.5	0.7	2.0	-0.4	20	100	15
-GATE3	-1.5	0.7	2.0	-0.4	20	100	15
-TRIG1	-1.5	0.7	2.0	-0.4	20	100	15
-TRIG2	-1.5	0.7	2.0	-0.4	20	100	15
-TRIG3	-1.5	0.7	2.0	-0.4	20	100	15

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
CLK CTR	-1.5	0.7	2.0	-2.4	100	400	25
IEI	-1.5	0.7	2.0	-0.74	-247	111	25
-IACK	-1.5	0.7	2.0	-0.74	-247	111	25
CLK	-1.5	0.8	2.0	-0.25	25	1000	50
-RST	-1.5	0.8	2.0	-0.25	25	1000	15
-CS	-1.5	0.8	2.0	-0.25	25	1000	15
-BES	-1.5	0.8	2.0	-0.25	25	1000	15
-EN	-1.5	0.8	2.0	-0.50	50	2000	70

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
DATA0	2.4	0.4	-3000	12.0	20	-200	-40	30
DATA1	2.4	0.4	-3000	12.0	20	-200	-40	30
DATA2	2.4	0.4	-3000	12.0	20	-200	-40	30
DATA3	2.4	0.4	-3000	12.0	20	-200	-40	30
DATA4	2.4	0.4	-3000	12.0	20	-200	-40	30
DATA5	2.4	0.4	-3000	12.0	20	-200	-40	30
DATA6	2.4	0.4	-3000	12.0	20	-200	-40	30
DATA7	2.4	0.4	-3000	12.0	20	-200	-40	30
PA0	2.4	0.5	-250	3.2	10	-10	NA	30
PA1	2.4	0.5	-250	3.2	10	-10	NA	30
PA2	2.4	0.5	-250	3.2	10	-10	NA	30
PA3	2.4	0.5	-250	3.2	10	-10	NA	30
PA4	2.4	0.5	-250	3.2	10	-10	NA	30
PA5	2.4	0.5	-250	3.2	10	-10	NA	30

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:06Z
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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
PA6	2.4	0.5	-250	3.2	10	-10	NA	30
PA7	2.4	0.5	-250	3.2	10	-10	NA	30
C/T1	2.4	0.5	-250	3.2	10	-10	NA	25
C/T2	2.4	0.5	-250	3.2	10	-10	NA	25
C/T3	2.4	0.5	-250	3.2	10	-10	NA	25
-INT	2.4	0.5	-250	3.2	NA	NA	NA	40
IEO	2.4	0.5	-225	2.95	NA	NA	NA	40
Q0	2.5	0.4	-400	4.0	NA	NA	-15	20
Q1	2.5	0.4	-400	4.0	NA	NA	-15	20
Q2	2.5	0.4	-400	4.0	NA	NA	-15	20
Q4	2.5	0.4	-400	4.0	NA	NA	-15	20
Q5	2.5	0.4	-400	4.0	NA	NA	-15	20
Q6	2.5	0.4	-400	4.0	NA	NA	-15	20

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
Q7	2.5	0.4	-400	4.0	NA	NA	-15	20
Q3	2.5	0.4	-300	1.6	NA	NA	-15	20
-ZAE	2.4	0.5	0.0	0.0	NA	NA	NA	85
-ZAS	2.4	0.5	0.0	0.0	NA	NA	NA	50
-ZDE	2.4	0.5	0.0	0.0	NA	NA	NA	60
-ZDS	2.4	0.5	0.0	0.0	NA	NA	NA	60
PU1	2.4	NA	-400	NA	NA	NA	NA	NA
PU2	2.4	NA	-2000	NA	NA	NA	NA	NA

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6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module function is printed in the appendix.

6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
DATA0	80	I/O	This 8-bit, bidirectional, three-state bus provides a means for data transfer to and from the CEM module.
DATA1	82	I/O	
DATA2	33	I/O	
DATA3	84	I/O	
DATA4	35	I/O	
DATA5	86	I/O	
DATA6	38	I/O	
DATA7	89	I/O	
ADDR0	85	I	These inputs provide addressing information to the CEM.
ADDR1	83	I	
ADDR2	74	I	
ADDR3	24	I	
ADDR4	27	I	
ADDR5	26	I	
ADDR6	81	I	
READ	32	I	This input determines the direction of the data transfer on the data bus. When high, data is read from the CEM. When low, data is written to the CEM.
CLK1	34	I	These inputs provide external clock inputs to the CEM counter/ timers 1, 2 and 3, respectively.
CLK2	36	I	
CLK3	88	I	
PCLK	37	I	This peripheral clock is the internal counter input used with the timers in the CEM.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-GATE1 -GATE2 -GATE3	05 92 42	I I I	These signals provide external gate inputs to the CEM counter/ timers 1, 2, and 3, respectively.
-TRIG1 -TRIG2 -TRIG3	06 93 43	I I I	Provides external trigger inputs to counter/timers 1, 2 and 3, respectively.
CLK	30	I	This input provides a clock input for the CEM timing sequence logic (PAL16R4).
-BES	69	I	This active low input signals the CEM that the data bus transfer is valid.
-CS	68	I	When active (low), this signal selects the CEM.
-RST	67	I	When active (low), this signal resets the CEM. The external gate and trigger input flip-flops and the binary counter are not affected by -RST.
-EN	09	I	When active (low), this signal enables the outputs of the Programmable Array Logic (PAL). This input is used for module testing only. It is grounded during normal operation.
CLK CTR	39	I	This input provides an input frequency to the independent binary counter logic (54LS393).

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
IEI	87	I	This active high input is used with IEO to form an interrupt daisy-chain when more than one CEM is a possible source of an interrupt. This signal has an internal module 10K ohm pull-up resistor which enables the IEI input when only one CEM module is used.
-IACK	29	I	This active low input signals to the CEM that the microprocessor is in an interrupt acknowledge cycle. This input has an internal module 10K ohm pull-up resistor which disables -IACK when the CEM is not used in the interrupt mode.
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7	61 62 13 64 65 16 19 20	O O O O O O O O	This is a general purpose CEM output port that can be used for information transfers. These outputs are in three-state mode after reset and until the CEM is properly programmed.
C/T1 C/T2 C/T3	14 11 22	O O O	These three-state outputs are the outputs of the 3 counter/timers of the CEM. The outputs are in three-state mode after reset and until the CEM is properly programmed active.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-INT	73	O	This active low output signals the microprocessor that the CEM requires interrupt service. The -INT output is open drain and can therefore be used in a wired OR configuration with other CEM modules. In the reset state the -INT output is not asserted.
IEO	21	O	This active high, three-state output is used with IEI when daisy-chaining multiple CEM modules. In the reset state the IEO output follows the IEI input pin.
-ZAE	08	I/O	This signal is used for test only. It gives the tester access to the Z8036 -CS0 input.
-ZDE	02	I/O	This signal is used for test only. It gives the tester access to the three-state enable input of the 54LS245 data transceiver.
-ZAS	58	I/O	This signal is used for test only. It gives the tester access to the Z8036 -AS input.
-ZDS	07	I/O	This signal is used for test only. It gives the tester access to the Z8036 -DS input.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
Q0	91	O	The independent binary counter provides outputs (Q0-Q7) that range in frequency from 23 KHz to 3 MHz, using 6 MHz as an input frequency (CLK CTR). Each output is less than the previous by a factor of two.
Q1	45	O	
Q2	95	O	
Q3	46	O	
Q4	94	O	
Q5	96	O	
Q6	97	O	
Q7	98	O	
PU1	23	O	These outputs are general purpose sources of logic 1 signals.
PU2	44	O	

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 7.

(j) Vol test - table XI, figure 7.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

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(n) Functional test - table XV, figure 8.

(o) Dynamic test - tables XVI and XVII, figures 9 and 10.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 11.

(r) Output crosstalk tests - tables XX and XXI, figures 12 and 13.

(s) Cin test - table XXII.

(t) Cout test - table XXIII.

(u) Threshold test - table XXIV.

(v) Disable time test - table XXV, figures 14 through 20.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 17, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0059)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused pins	All used pins	All unused pins	10	INF	MOhm	1-7
A3	Isolation, unused to unused pins	100	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70,71,72, 75,76,77, 78,79,90, 99	10	INF	MOhm	1-7
A4	Isolation, unused to unused pins	99	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70,71,72, 75,76,77, 78,79,90,	10	INF	MOhm	1-7
A5	Isolation, unused to unused pins	90	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70,71,72, 75,76,77, 78,79	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A6	Isolation, unused to unused pins	79	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70,71,72, 75,76,77, 78	10	INF	MOhm	1-7
A7	Isolation, unused to unused pins	78	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70,71,72, 75,76,77,	10	INF	MOhm	1-7
A8	Isolation, unused to unused pins	77	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70,71,72, 75,76	10	INF	MOhm	1-7
A9	Isolation, unused to unused pins	76	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70,71,72, 75	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A10	Isolation, unused to unused pins	75	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70,71,72,	10	INF	MOhm	1-7
A11	Isolation, unused to unused pins	72	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70,71	10	INF	MOhm	1-7
A12	Isolation, unused to unused pins	71	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66, 70	10	INF	MOhm	1-7
A13	Isolation, unused to unused pins	70	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63,66	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A14	Isolation, unused to unused pins	66	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60,63	10	INF	MOhm	1-7
A15	Isolation, unused to unused pins	63	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59, 60	10	INF	MOhm	1-7
A16	Isolation, unused to unused pins	60	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57,59	10	INF	MOhm	1-7
A17	Isolation, unused to unused pins	59	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56,57	10	INF	MOhm	1-7
A18	Isolation, unused to unused pins	57	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55, 56	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A19	Isolation, unused to unused pins	56	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54,55	10	INF	MOhm	1-7
A20	Isolation, unused to unused pins	55	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53,54	10	INF	MOhm	1-7
A21	Isolation, unused to unused pins	54	3,4,12,17, 18,28,41, 47,48,49, 50,51,52, 53	10	INF	MOhm	1-7
A22	Isolation, unused to unused pins	53	3,4,12,17, 18,28,41, 47,48,49, 50,51,52,	10	INF	MOhm	1-7
A23	Isolation, unused to unused pins	52	3,4,12,17, 18,28,41, 47,48,49, 50,51	10	INF	MOhm	1-7
A24	Isolation, unused to unused pins	51	3,4,12,17, 18,28,41, 47,48,49, 50	10	INF	MOhm	1-7
A25	Isolation, unused to unused pins	50	3,4,12,17, 18,28,41, 47,48,49	10	INF	MOhm	1-7
A26	Isolation, unused to unused pins	49	3,4,12,17, 18,28,41, 47,48	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A27	Isolation, unused to unused pins	48	3,4,12,17, 18,28,41, 47	10	INF	MOhm	1-7
A28	Isolation, unused to unused pins	47	3,4,12,17, 18,28,41	10	INF	MOhm	1-7
A29	Isolation, unused to unused pins	41	3,4,12,17, 18,28	10	INF	MOhm	1-7
A30	Isolation, unused to unused pins	28	3,4,12,17, 18	10	INF	MOhm	1-7
A31	Isolation, unused to unused pins	18	3,4,12,17,	10	INF	MOhm	1-7
A32	Isolation, unused to unused pins	17	3,4,12	10	INF	MOhm	1-7
A33	Isolation, unused to unused pins	12	3,4	10	INF	MOhm	1-7
A34	Isolation, unused to unused pins	4	3	10	INF	MOhm	1-7

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V 30 V	± 10 percent ± 10 percent	A1 A2-A34
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A1-A34 A1-A34

2. The megohm bridge used shall be item R06.

3. In the setup conditions of table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	01	31	0.0	1.0	Ohm	1-4
B2	Continuity, GND	10	15				
B3	Continuity, GND	15	25				
B4	Continuity, GND	25	40	0.0	1.0	Ohm	1-4
B5	Continuity, Vcc	01	31	0.0	0.1	Ohm	1-4
B6	Continuity, GND	10	15				
B7	Continuity, GND	15	25				
B8	Continuity, GND	25	40	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B4 B5-B8

2. The ohmmeter used shall be item R10.

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NOTES: - Continued.

3. In the setup conditions of table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.044 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	ADDR0	85	5.5	0.0	0.1	mA	1-3
D2	ADDR1	83					
D3	ADDR2	74					
D4	ADDR3	24					
D5	ADDR4	27					
D6	ADDR5	26					
D7	ADDR6	81					
D8	READ	32					
D9	CLK1	34					
D10	CLK2	36					
D11	CLK3	88					
D12	PCLK	37					
D13	DATA0	80					
D14	DATA1	82					
D15	DATA2	33					
D16	DATA3	84					
D17	DATA4	35					
D18	DATA5	86					
D19	DATA6	38					
D20	DATA7	89	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	-GATE1	05	5.5	0.0	0.1	mA	1-3
D22	-GATE2	92					
D23	-GATE3	42					
D24	-TRIG1	06					
D25	-TRIG2	93					
D26	-TRIG3	43	5.5	0.0	0.1	mA	1-3
D27	CLK	30	5.5	0.0	1.0	mA	1-3
D28	-BES	69					
D29	-CS	68					
D30	-RST	67	5.5	0.0	1.0	mA	1-3
D31	-EN	09	5.5	0.0	2.0	mA	1-3
D32	CLK CTR	39	5.5	0.0	0.4	mA	1-3
D33	IEI	87	5.5	0.0	0.111	mA	1-3
D34	-IACK	29	5.5	0.0	0.111	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V 5.5 V	± 55 mV + 0mV, -55mV	D1-D32 D33,D34
Power supply current limit	1.044 A	NA	D1-D34
Forcing voltage, (Vin)	5.5 V 5.5 V	± 55 mV -0 mV,+55 mV	D1-D32 D33,D34
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	D1-D34 D1-D34
Input current measurement	NA NA NA NA NA	± 5.0 uA ± 50.0 uA ± 100.0 uA ± 20.0 uA ± 5.6 uA	D1-D26 D27-D30 D31 D32 D33,D34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	ADDR0	85	-18.0	-1.5	-0.4	V	1-3
E2	ADDR1	83					
E3	ADDR2	74					
E4	ADDR3	24					
E5	ADDR4	27					
E6	ADDR5	26					
E7	ADDR6	81					
E8	READ	32					
E9	CLK1	34					
E10	CLK2	36					
E11	CLK3	88					
E12	PCLK	37					
E13	DATA0	80					
E14	DATA1	82					
E15	DATA2	33					
E16	DATA3	84					
E17	DATA4	35					
E18	DATA5	86					
E19	DATA6	38					
E20	DATA7	89	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E21	-GATE1	05	-18.0	-1.5	-0.4	V	1-3
E22	-GATE2	92					
E23	-GATE3	42					
E24	-TRIG1	06					
E25	-TRIG2	93					
E26	-TRIG3	43					
E27	CLK	30					
E28	-BES	69					
E29	-CS	68					
E30	-RST	67					
E31	-EN	09					
E32	CLK CTR	39					
E33	IEI	87					
E34	-IACK	29	-18.0	-1.5	-0.4	V	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:06Z
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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E34
Power supply current limit	1.044 A	NA	E1-E34
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E34
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E34
low, (VL)	0.0 V	± 100 mV	E1-E34
Input voltage measurement	NA	± 55 mV	E1-E34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	Q0	91	0.0	-100	-15	mA	1-4
F2	Q1	45					
F3	Q2	95					
F4	Q3	46					
F5	Q4	94					
F6	Q5	96					
F7	Q6	97					
F8	Q7	98	0.0	-100	-15	mA	1-4
F9	DATA0	80	0.0	-225	-40	mA	1-4
F10	DATA1	82					
F11	DATA2	33					
F12	DATA3	84					
F13	DATA4	35					
F14	DATA5	86					
F15	DATA6	38					
F16	DATA7	89	0.0	-225	-40	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:06Z
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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F16
Power supply current limit	1.044 A	NA	F1-F16
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F16
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F16 F1-F16
Output current measurement	 NA NA	 ± 4.25 mA ± 9.25 mA	 F1-F8 F9-F16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:06Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	ADDR0	85	2.7	0	20	uA	1-3
G2	ADDR1	83					
G3	ADDR2	74					
G4	ADDR3	24					
G5	ADDR4	27					
G6	ADDR5	26					
G7	ADDR6	81					
G8	READ	32					
G9	CLK1	34					
G10	CLK2	36					
G11	CLK3	88					
G12	PCLK	37					
G13	DATA0	80					
G14	DATA1	82					
G15	DATA2	33					
G16	DATA3	84					
G17	DATA4	35					
G18	DATA5	86					
G19	DATA6	38					
G20	DATA7	89	2.7	0	20	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	-GATE1	05	2.7	0	20	uA	1-3
G22	-GATE2	92					
G23	-GATE3	42					
G24	-TRIG1	06					
G25	-TRIG2	93					
G26	-TRIG3	43	2.7	0	20	uA	1-3
G27	IEI	87	2.7	-294	-247	uA	1-3
G28	-IACK	29	2.7	-294	-247	uA	1-3
G29	CLK CTR	39	2.7	-150	100	uA	1-3
G30	CLK	30	2.4	0	25	uA	1-3
G31	-RST	67					
G32	-CS	68					
G33	-BES	69	2.4	0	25	uA	1-3
G34	-EN	09	2.4	0	50	uA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G34
Power supply current limit	1.044 A	NA	G1-G34
Forcing voltage, (Vih)	2.7 V 2.4 V	± 135 mV ± 120 mV	G1-G29 G30-G34
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G34 G1-G34
Input current measurement	NA NA NA NA NA	± 1.0 uA ± 2.35 uA ± 12.5 uA ± 1.25 uA ± 2.5 uA	G1-G26 G27,G28 G29 G30-G33 G34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	ADDR0	85	0.4	-200	0	uA	1-3
H2	ADDR1	83					
H3	ADDR2	74					
H4	ADDR3	24					
H5	ADDR4	27					
H6	ADDR5	26					
H7	ADDR6	81					
H8	CLK1	34					
H9	CLK2	36					
H10	CLK3	88					
H11	PCLK	37					
H12	READ	32					
H13	DATA0	80					
H14	DATA1	82					
H15	DATA2	33					
H16	DATA3	84					
H17	DATA4	35					
H18	DATA5	86					
H19	DATA6	38					
H20	DATA7	89	0.4	-200	0	uA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	-GATE1	05	0.4	-400	-30	uA	1-3
H22	-GATE2	92					
H23	-GATE3	42					
H24	-TRIG1	06					
H25	-TRIG2	93					
H26	-TRIG3	43	0.4	-400	-30	uA	1-3
H27	CLK	30	0.4	-250	0	uA	1-3
H28	-RST	67					
H29	-CS	68					
H30	-BES	69	0.4	-250	0	uA	1-3
H31	-EN	09	0.4	-500	0	uA	1-3
H32	IEI	87	0.4	-742	-480	uA	1-3
H33	-IACK	29	0.4	-742	-480	uA	1-3
H34	CLK CTR	39	0.4	-2400	-350	uA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H34
Power supply current limit	1.044 A	NA	H1-H34
Forcing voltage, (Vil)	0.4 V	± 20 mV	H1-H34
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H34
low, (VL)	0.0 V	± 100 mV	H1-H34
Input current measurement	NA	± 10.0 uA	H1-H20
	NA	± 18.5 uA	H21-H26
	NA	± 12.5 uA	H27-H30
	NA	± 25.0 uA	H31
	NA	± 13.1 uA	H32,H33
	NA	± 102.5 uA	H34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:06Z
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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	DATA0	80	-3.0	2.4	4.5	V	1-4
J2	DATA1	82					
J3	DATA2	33					
J4	DATA3	84					
J5	DATA4	35					
J6	DATA5	86					
J7	DATA6	38					
J8	DATA7	89	-3.0	2.4	4.5	V	1-4
J9	PA0	61	-0.25	2.4	4.5	V	1-4
J10	PA1	62					
J11	PA2	13					
J12	PA3	64					
J13	PA4	65					
J14	PA5	16					
J15	PA6	19					
J16	PA7	20					
J17	C/T1	14					
J18	C/T2	11					
J19	C/T3	22					
J20	-INT	73	-0.25	2.4	4.5	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	IEO	21	-0.225	2.4	4.5	V	1-4
J22	-ZAE	08	0.0	2.4	4.5	V	1-4
J23	-ZDE	02					
J24	-ZAS	58					
J25	-ZDS	07	0.0	2.4	4.5	V	1-4
J26	Q0	91	-0.4	2.5	4.5	V	1-4
J27	Q1	45					
J28	Q2	95					
J29	Q4	94					
J30	Q5	96					
J31	Q6	97					
J32	Q7	98	-0.4	2.5	4.5	V	1-4
J33	Q3	46	-0.3	2.5	4.5	V	1-4
J34	PU1	23	-0.4	2.4	2.6	V	1-4
J35	PU2	44	-2.0	2.4	2.6	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:06Z
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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J35
Power supply current limit	1.044 A	NA	J1-J35
Forcing current, (Ioh)	-3.0 mA	± 30.0 uA	J1-J8
	-0.25 mA	± 2.5 uA	J9-J20
	-0.225 mA	± 2.25 uA	J21
	0.0 mA	± 1.0 uA	J22-J25
	-0.4 mA	± 4.0 uA	J26-J32
	-0.3 mA	± 3.0 uA	J33
	-0.4 mA	± 4.0 uA	J34
	-2.0 mA	± 20.0 uA	J35
Pattern voltage high, (VH)	2.0 V	± 20 mV	J1-J35
low, (VL)	0.7 V	± 7 mV	J1-J35
Output voltage measurement	NA	± 105 mV	J1-J25
	NA	± 100 mV	J26-J33
	NA	± 10 mV	J34,J35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:06Z
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NOTES: - Continued.

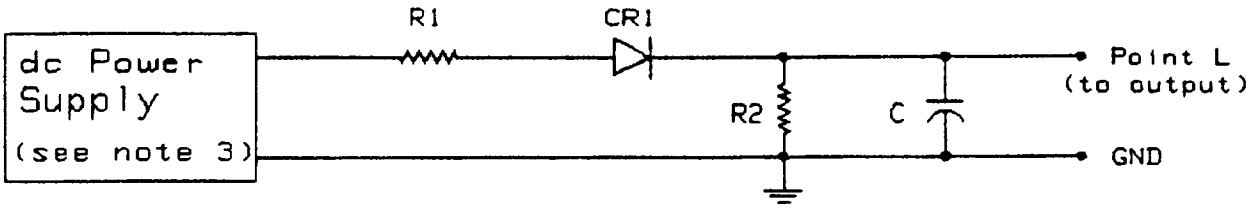
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 7.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

3. For the open drain outputs the load power supply shall be 5.0 V ± 1 percent, CR1, and R2 shall be deleted, and R1 shall be 1406.25 ohms.

FIGURE 7. Logic load circuit.

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NOTES: - Continued.

4. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time tests, 5.0 V ± 1 percent for the open drain outputs (see note 3) and 2.5 V ± 1 percent for all other tests. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are in the table below.

5. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
61,62,13,64,65, 16,19,20,14,11,22	9600	3.2	0.5
91,45,94,95,96, 97,98	6250	4.0	0.4
80,82,33,84,35, 86,38,89	800	12.0	0.4
46	8333	1.6	0.4
21	10666	2.95	0.5
73	NA	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	DATA0	80	12.0	0.0	0.4	V	1-4
K2	DATA1	82					
K3	DATA2	33					
K4	DATA3	84					
K5	DATA4	35					
K6	DATA5	86					
K7	DATA6	38					
K8	DATA7	89	12.0	0.0	0.4	V	1-4
K9	Q0	91	4.0	0.0	0.4	V	1-4
K10	Q1	45					
K11	Q2	95					
K12	Q4	94					
K13	Q5	96					
K14	Q6	97					
K15	Q7	98	4.0	0.0	0.4	V	1-4
K16	Q3	46	1.6	0.0	0.4	V	1-4
K17	PA0	61	3.2	0.0	0.5	V	1-4
K18	PA1	62					
K19	PA2	13					
K20	PA3	64	3.2	0.0	0.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	PA4	65	3.2	0.0	0.5	V	1-4
K22	PA5	16					
K23	PA6	19					
K24	PA7	20					
K25	C/T1	14					
K26	C/T2	11					
K27	C/T3	22					
K28	-INT	73	3.2	0.0	0.5	V	1-4
K29	IEO	21	2.95	0.0	0.5	V	1-4
K30	-ZAE	08	0.0	0.0	0.5	V	1-4
K31	-ZDE	02					
K32	-ZAS	58					
K33	-ZDS	07	0.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:06Z
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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K33
Power supply current limit	1.044 A	NA	K1-K33
Forcing current, (Iol)	12.0 mA 4.0 mA 1.6 mA 3.2 mA 2.95 mA 0.0 mA	± 120.0 uA ± 40.0 uA ± 16.0 uA ± 32.0 uA ± 29.5 uA ± 1.0 uA	K1-K8 K9-K15 K16 K17-K28 K29 K30-K33
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.7 V	 ± 20 mV ± 7 mV	 K1-K33 K1-K33
Output voltage measurement	NA NA	± 20 mV ± 25 mV	K1-K16 K17-K33

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 7.

4. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	DATA0	80	2.7	-20	20	uA	1-3
L2	DATA1	82					
L3	DATA2	33					
L4	DATA3	84					
L5	DATA4	35					
L6	DATA5	86					
L7	DATA6	38					
L8	DATA7	89	2.7	-20	20	uA	1-3
L9	PA0	61	2.4	-10	10	uA	1-3
L10	PA1	62					
L11	PA2	13					
L12	PA3	64					
L13	PA4	65					
L14	PA5	16					
L15	PA6	19					
L16	PA7	20					
L17	C/T1	14					
L18	C/T2	11					
L19	C/T3	22	2.4	-10	10	uA	1-3

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L19
Power supply current limit	1.044 A	NA	L1-L19
Forcing voltage, (Voh)	2.7 V 2.4 V	± 135 mV ± 120 mV	L1-L8 L9-L19
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L19 L1-L19
Output current measurement	NA NA	± 2 uA ± 1 uA	L1-L8 L9-L19

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	DATA0	80	0.4	-200	20	uA	1-3
M2	DATA1	82					
M3	DATA2	33					
M4	DATA3	84					
M5	DATA4	35					
M6	DATA5	86					
M7	DATA6	38					
M8	DATA7	89	0.4	-200	20	uA	1-3
M9	PA0	61	0.4	-10	10	uA	1-3
M10	PA1	62					
M11	PA2	13					
M12	PA3	64					
M13	PA4	65					
M14	PA5	16					
M15	PA6	19					
M16	PA7	20					
M17	C/T1	14					
M18	C/T2	11					
M19	C/T3	22	0.4	-10	10	uA	1-3

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NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M19
Power supply current limit	1.044 A	NA	M1-M19
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M19
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	M1-M19 M1-M19
Output current measurement	NA NA	± 11 uA ± 1 uA	M1-M8 M9-M19

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions of table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	70	696	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.044 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 31.3 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.044 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 150 pF (see figure 7). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

3. Abbreviations used are defined in 6.1.

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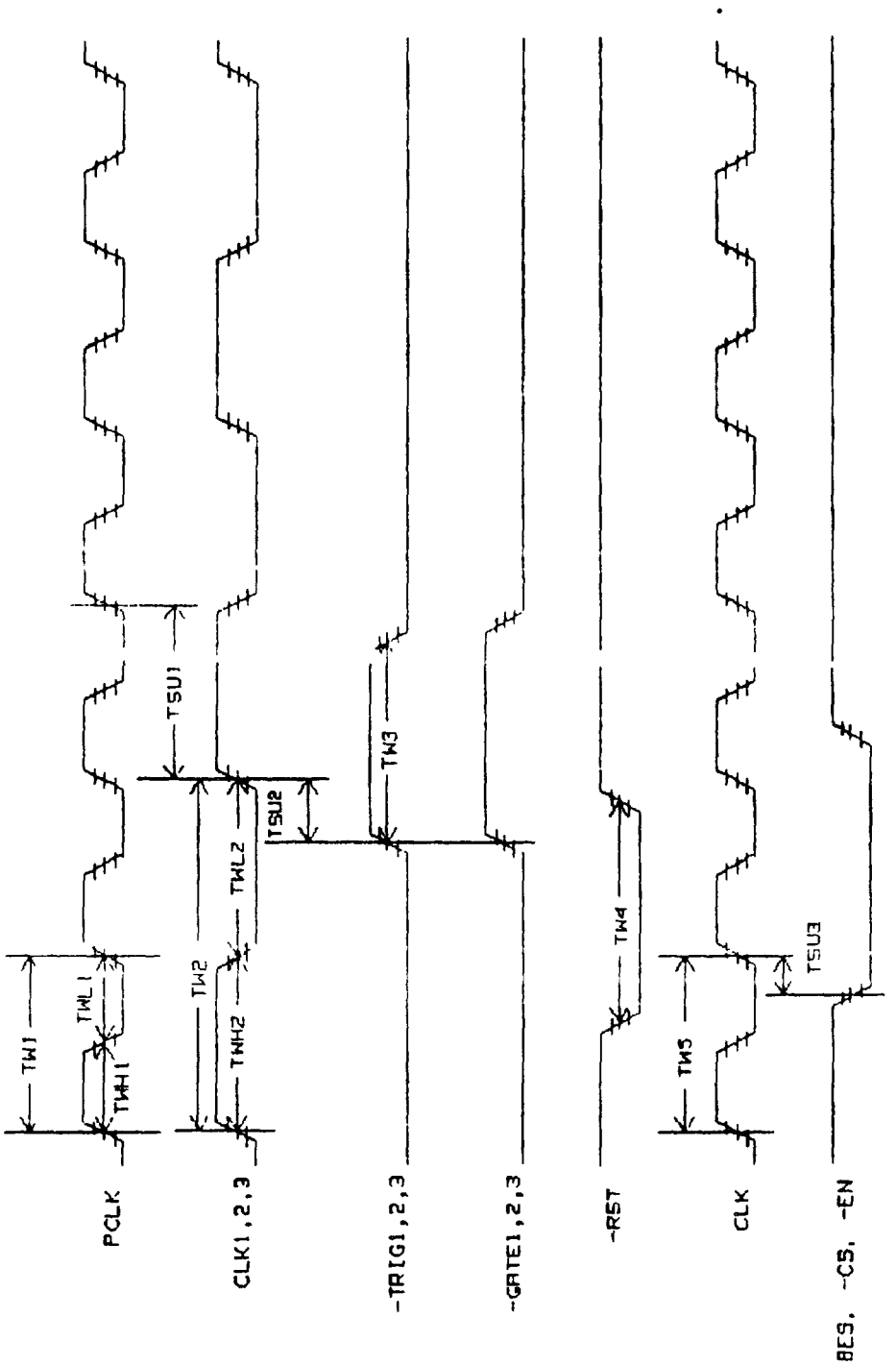


FIGURE 8. Input timing parameters for table XV.

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NOTES:

1. Input parameters for figure 8:

TW1 = 333 nS minimum.
TWH1 = TWL1 = 126 nS minimum.
TW2 = 666 nS minimum.
TWH2 = TWL2 = 251 nS minimum.
TW3 = 254 nS minimum.
TW4 = 300 nS minimum.
TW5 = 200 nS minimum.
TSU1 = 210 nS minimum.
TSU2 = 20 nS minimum.
TSU3 = 30 nS minimum.

2. All inputs shall have rise and fall times less than or equal to 12 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-15
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-15
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-15
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-15

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.044 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 200 mV ± 35 mV	Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1,Q3
high, (Voh)	NA	± 155 mV	Q2,Q4
low, (Vol)	NA	± 25 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4 \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4$

3. The load circuit for this test is specified on figure 7.

4. The output pin shall be loaded as shown on figure 7, except C = 300 pF ± 30 pF.

5. All of the input waveforms shall meet the requirements shown on figure 9.

6. Input/Output (I/O) waveforms are as referenced on figure 10.

7. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

8. T = CLK clock period.

9. The respective input must also meet the setup time for the dual positive-edge-triggered D flip-flop (parameters 58-60) for this parameter to be valid.

10. -BES must also be low during the cycle.

11. Required time for valid reset of Z8036.

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NOTES: - Continued.

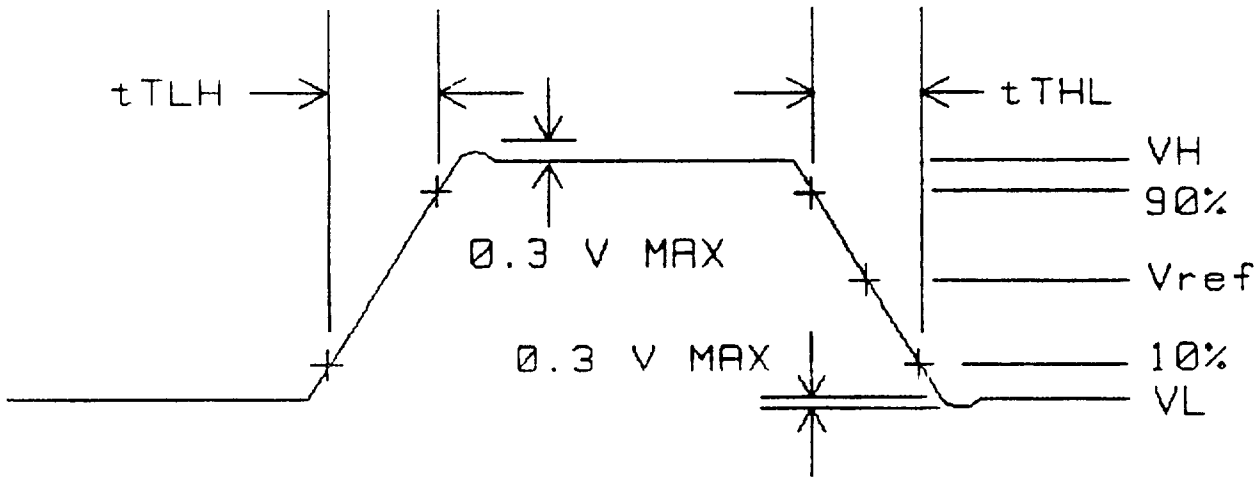
12. The timing requirements of 25 and 28 (figure 10c) are required for the counter mode. The counter starts on the rising edge of the counter input.

13. The timing requirements indicated by references 25, 26 and 28 are required for timer mode. Counter starts on the rising edge of PCLK/2.

14. PCLK/2 is an internal signal that is shown for illustrative information only.

15. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.3 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
39,85,83,74,24,27,26,81,80, 82,33,84,35,86,38,89,30,67, 68,69,9,32,34,36,88,37,87,29, 5,6,92,93,42,43	15 ± 1.5 nS	6 ± 1.5 nS

FIGURE 9. Sample input waveform for table XVI.

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TABLE XVII Dynamic parameter table.

Parameter number	Parameter	0 to 60°C		25 ± 5°C		Units	Fig	Ref
		Min	Max	Min	Max			
1	Setup time from -BES to CLK	30	-	30	-	nS	10a 10b 10d	1
2	Setup time from -CS to CLK	30	-	30	-	nS	10a 10b	2
3	Setup time from READ to CLK	81	-	63	-	nS	10a 10b 10d	11
4-10	Setup time from ADDR0 -ADDR6 to CLK (address setup	53	-	48	-	nS	10a 10b	12
11-17	Hold time from CLK to ADDR0-ADDR6 (address hold)	98	-	98	-	nS	10a 10b	13
18-25	Setup time from DATA0-DATA7 to CLK (data setup-write cycle)	52	-	48	-	ns	10b	14
26-33	Hold time from CLK to DATA0-DATA7 invalid (data hold-write cycle)	55	-	55	-	nS	10b	15
34-41	Delay time from CLK(RE) to DATA0-DATA7 valid (read cycle)	-	339	-	335	nS	10a 10b 10d	16
42	CLK clock period	200	2000	200	2000	nS	10a 10b 10d	18

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TABLE XVII Dynamic parameter table - Continued.

Parameter number	Parameter	0 to 60°C		25 ± 5°C		Units	Fig	Ref.
		Min	Max	Min	Max			
43	PCLK cycle time	333	3980	333	3984	nS	10c	19
44	PCLK high width	126	1979	121	1984	nS	10c	20
45	PCLK low width	126	1979	121	1984	nS	10c	21
46-48	Counter inputs (CLK1-CLK3) cycle time	666	-	666	-	nS	10c	22
49-51	Counter inputs (CLK1-CLK3) low width	251	-	246	-	nS	10c	23
52-54	Counter inputs (CLK1-CLK3) high width	251	-	246	-	nS	10c	24
55-57	Trigger inputs(-TRIG1, 2,3) to counter inputs (CLK1,2,3) setup time (counter mode)	20	-	20	-	nS	10c	25
58-60	Counter inputs (CLK1,2, 3) to PCLK(FE) setup time	210	-	197	-	nS	10c	26
61-63	Trigger inputs (-TRIG1, 2,3) to counter inputs (CLK1,2,3) hold time	25	-	25	-	nS	10c	27

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TABLE XVII. Dynamic parameter table - Continued.

Parameter number	Parameter	0 to 60°C		25 ± 5°C		Units	Fig	Ref
		Min	Max	Min	Max			
64-66	Gate inputs(-GATE1,2,3) to counter inputs (CLK1,2,3) setup time (counter mode)	20	-	20	-	nS	10c	28
67-69	Gate inputs (-GATE1,2, 3) to counter input (CLK1,2,3) hold time	25	-	25	-	nS	10c	29
70-72	Counter inputs (CLK1,2, 3) to counter outputs (C/T1,C/T2,C/T3) delay (counter mode)	-	498	-	493	nS	10c	30
73-75	PCLK Counter outputs (C/T1,C/T2,C/T3) delay (timer mode)	-	498	-	493	nS	10c	31
76	-IACK setup time to CLK	52	-	48	-	nS	10d	32
77	-IACK hold time from CLK	298	-	298	-	nS	10d	33
78	Delay from CLK(RE) to IEO low (interrupt acknowledge cycle)	-	406	-	406	nS	10d	34
79	Delay from CLK(RE) to -INT high (interrupt acknowledge cycle)	-	661	-	661	nS	10d	35

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TABLE XVII. Dynamic parameter table - Continued.

Parameter number	Parameter	0 to 60°C		25 ± 5°C		Units	Fig	Ref
		Min	Max	Min	Max			
80-87	Delay from CLK(RE) to DATA0-DATA7 (read data) valid (interrupt acknowledge cycle)	-	339	-	335	nS	10d	36
88	Pattern match (port A) to -INT low delay.		8T+ 884		8T+ 880	nS	10e	37
89	CLK(RE) to -INT low (software generated interrupt)		368		368	nS	10e	38
90-92	Counter inputs (CLK1-CLK3) to -INT low delay (top count)	-	8T+ 784	-	8T+ 780	nS	10e	39
93	PCLK to -INT low delay (timer mode)	-	8T+ 784	-	8T+ 780	nS	10e	39
94	CLK CTR(FE) to O0 low (tPHL)	-	45	-	37	nS	10f	40
95	CLK CTR(FE) to O0 high (tPLH)	-	45	-	37	nS	10f	41
96	CLK CTR(FE) to O3 low (tPHL)	-	132	-	113	nS	10f	42
97	CLK CTR(FE) to O3 high (tPLH)	-	132	-	113	nS	10f	43

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TABLE XVII Dynamic parameter table - Continued.

Parameter number	Parameter	0 to 60°C		25 ± 5°C		Units	Fig	Ref
		Min	Max	Min	Max			
98	-RST pulse width low (note 11)	300	-	300	-	nS	10g	44
99	Setup times -EN, to CLK	30	-	30	-	nS	10g	45
100	Hold times CLK to -EN	0	-	0	-	nS	10g	46
101	IEI setup time to CLK	121	-	116	-	nS	10g	47
102	IEI hold time from CLK	153	-	153	-	nS	10g	48
103-110	Delay time from CLK(RE) to DATA0-DATA7 valid (Read cycle) (note 4)	-	353	-	349	nS	10a	16
111-118	Delay from CLK (RES to DATA0-DATA7 (read data) out valid (interrupt acknowledge cycle) (note 4)	-	353	-	349	nS	10d	36

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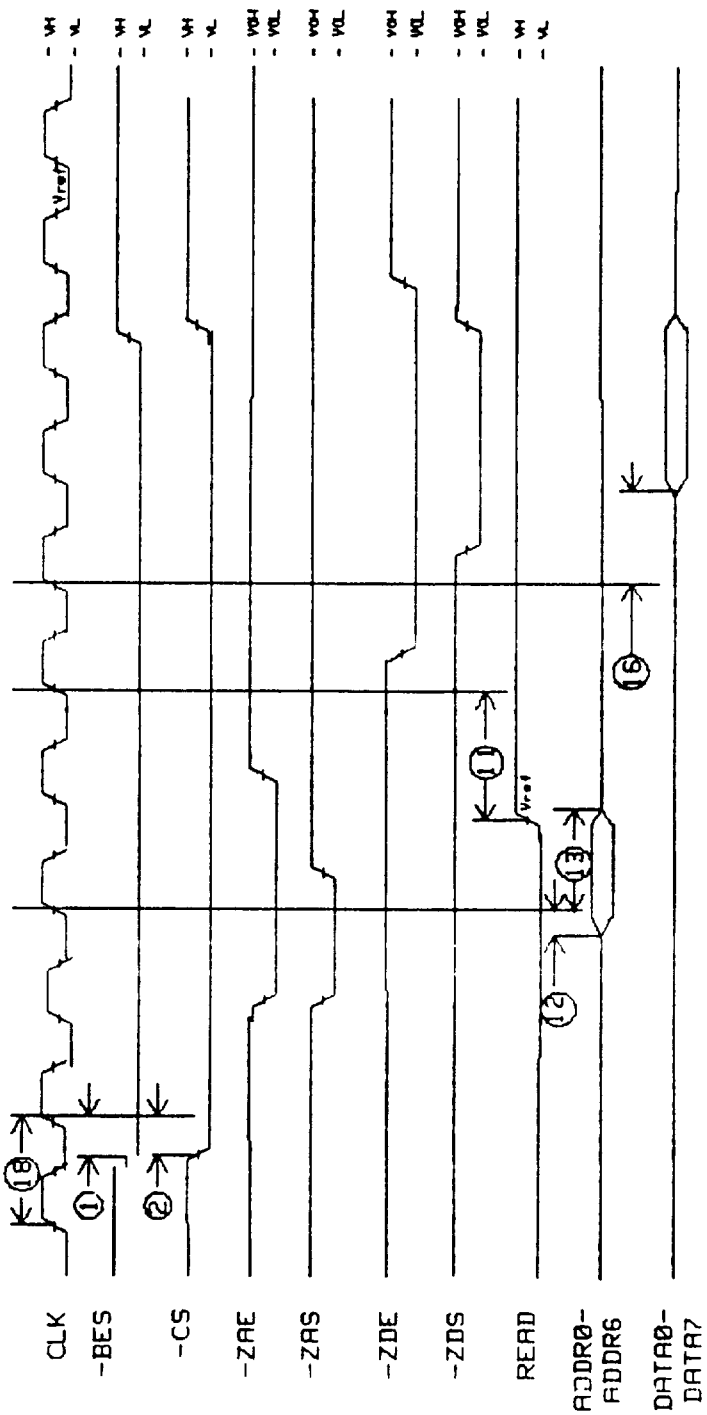


FIGURE 10a. Dynamic waveforms for tables XVI and XVII.

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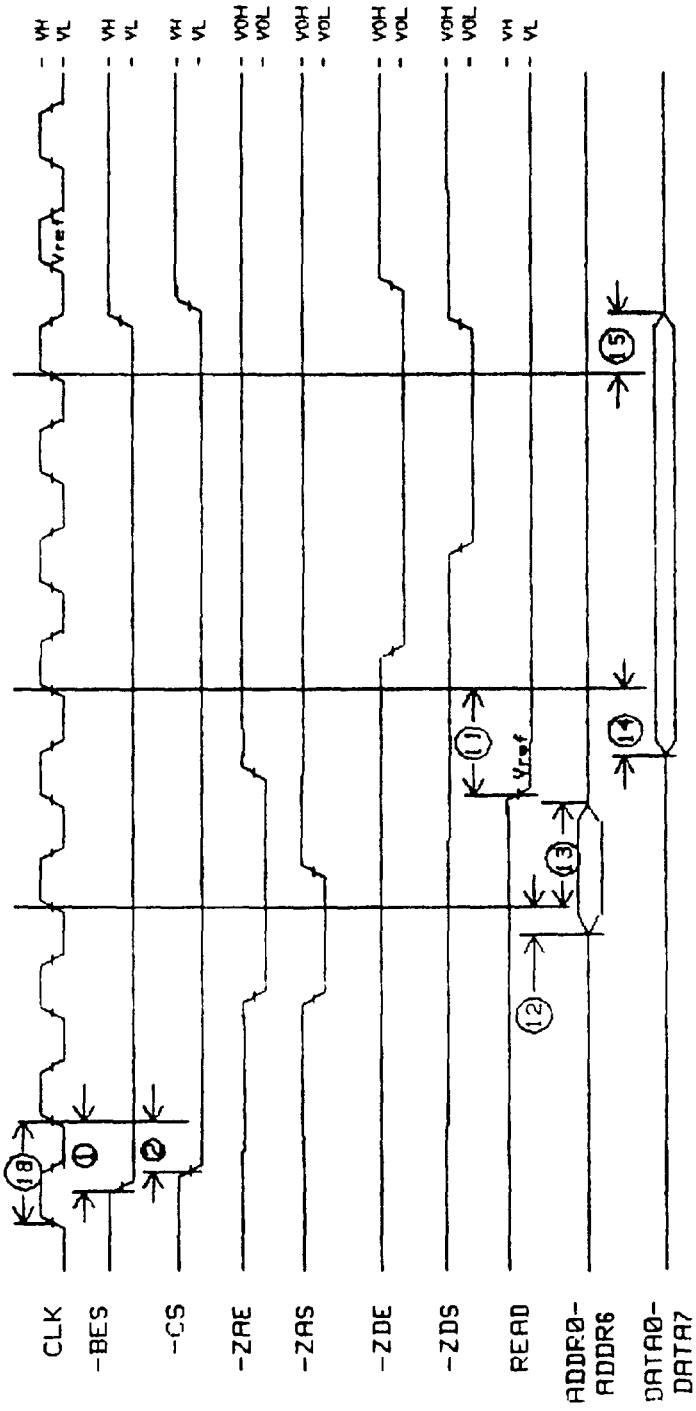


FIGURE 10b. Dynamic waveforms for tables XVI and XVII.

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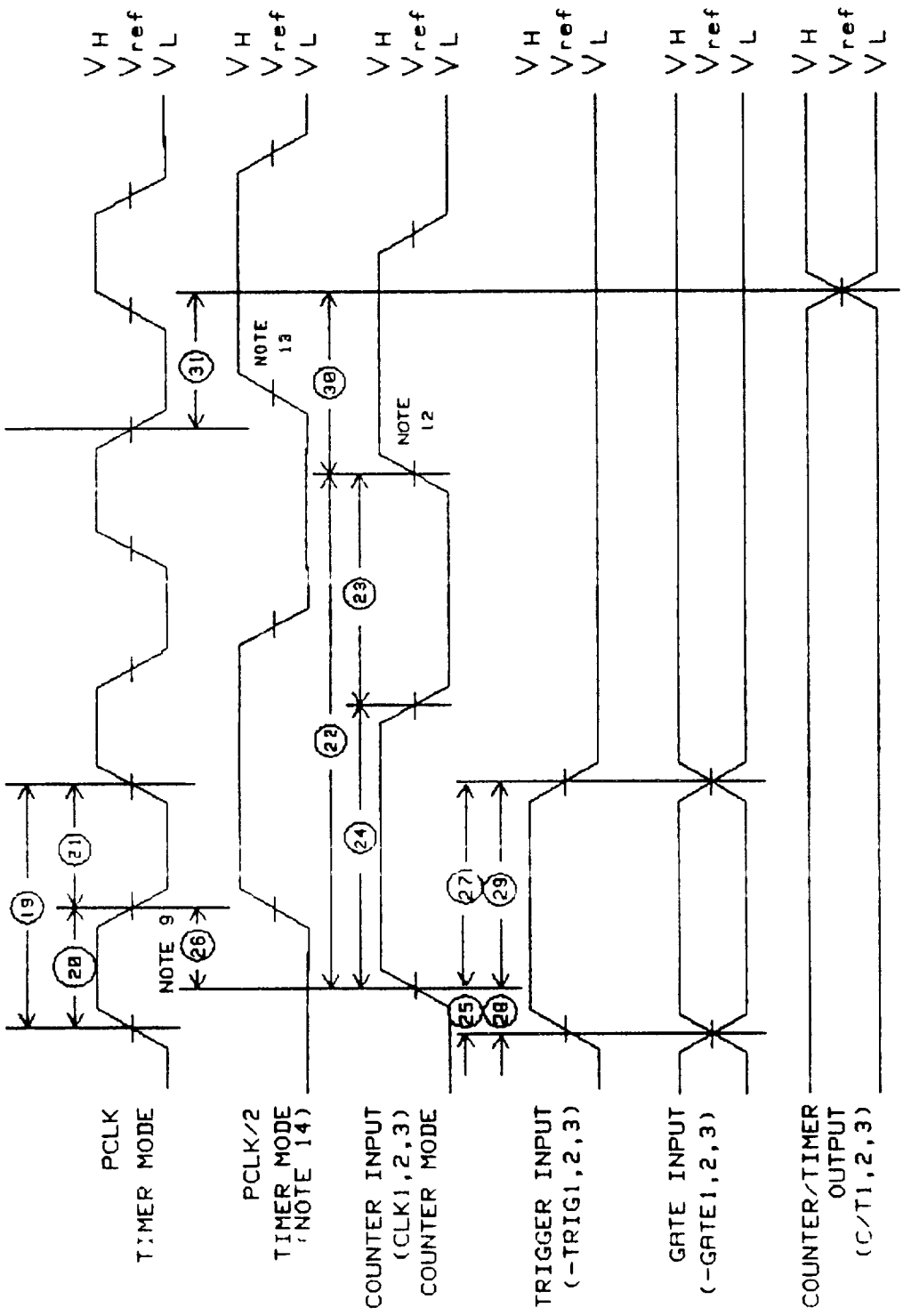


FIGURE 10c. Dynamic waveforms for tables XVI and XVII.

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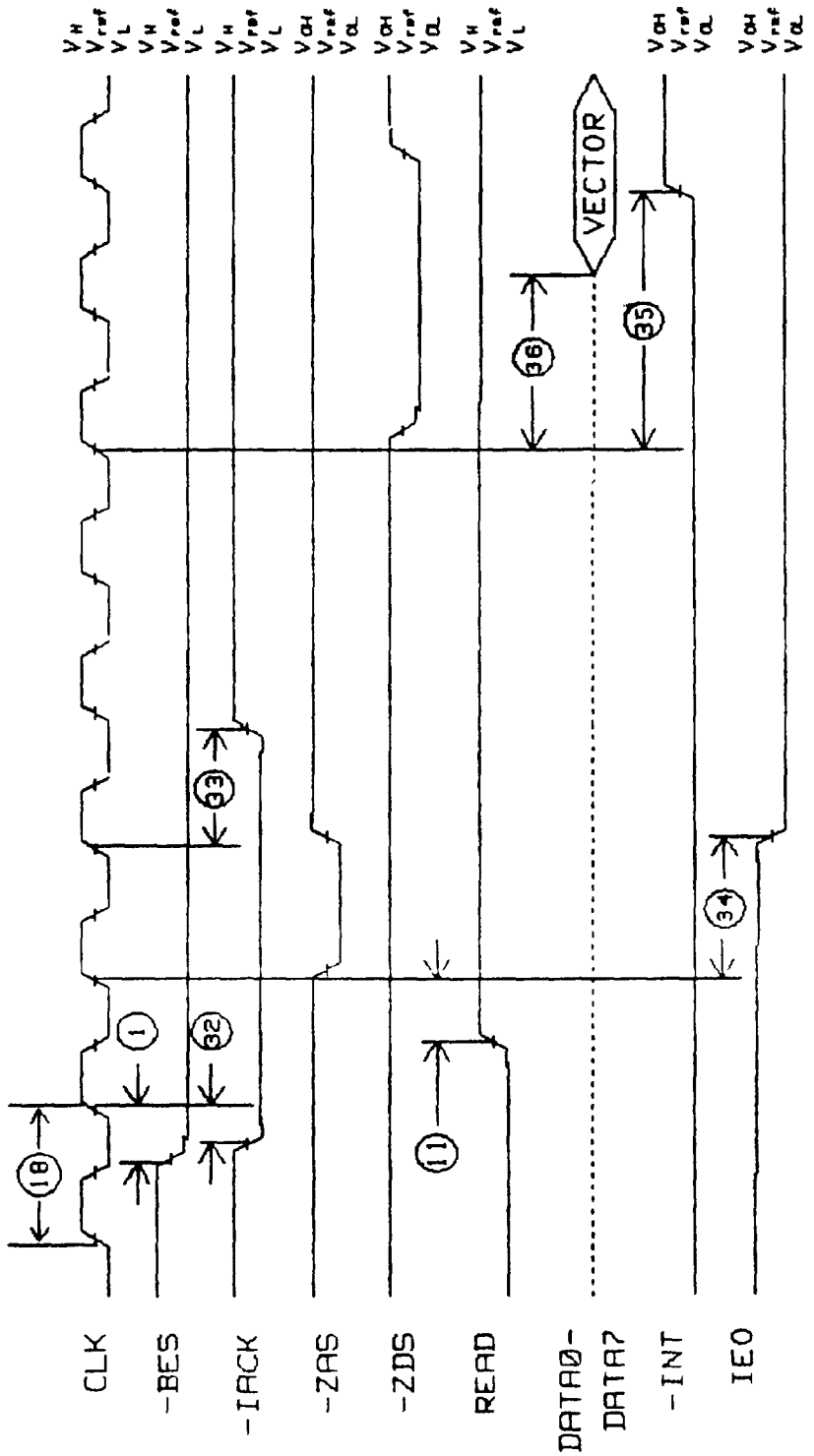


FIGURE 10d. Dynamic waveforms for tables XVI and XVII.

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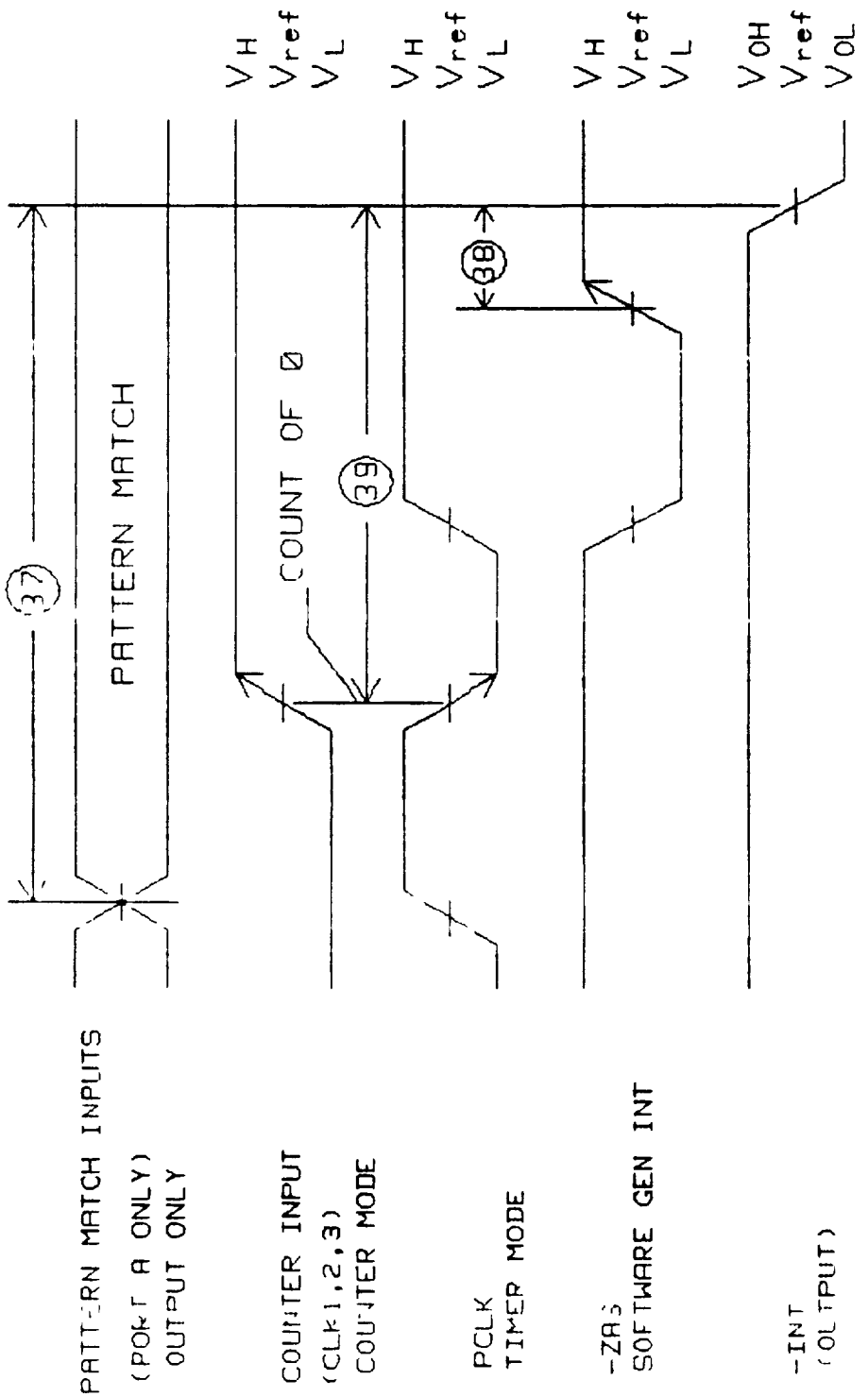


FIGURE 10e. Dynamic waveforms for tables XVI and XVII.

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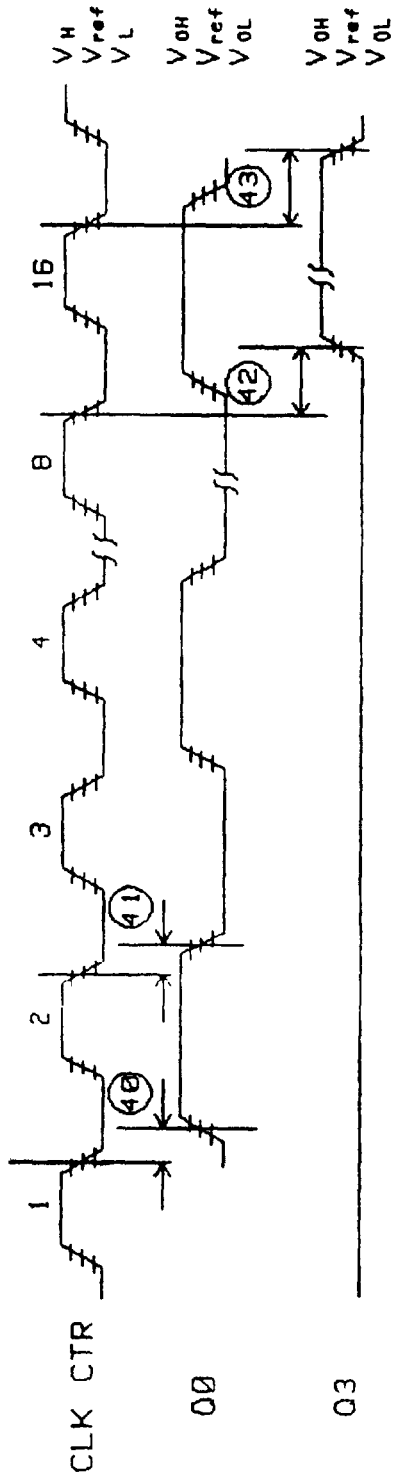


FIGURE 10f. Dynamic waveforms for tables XVI and XVII.

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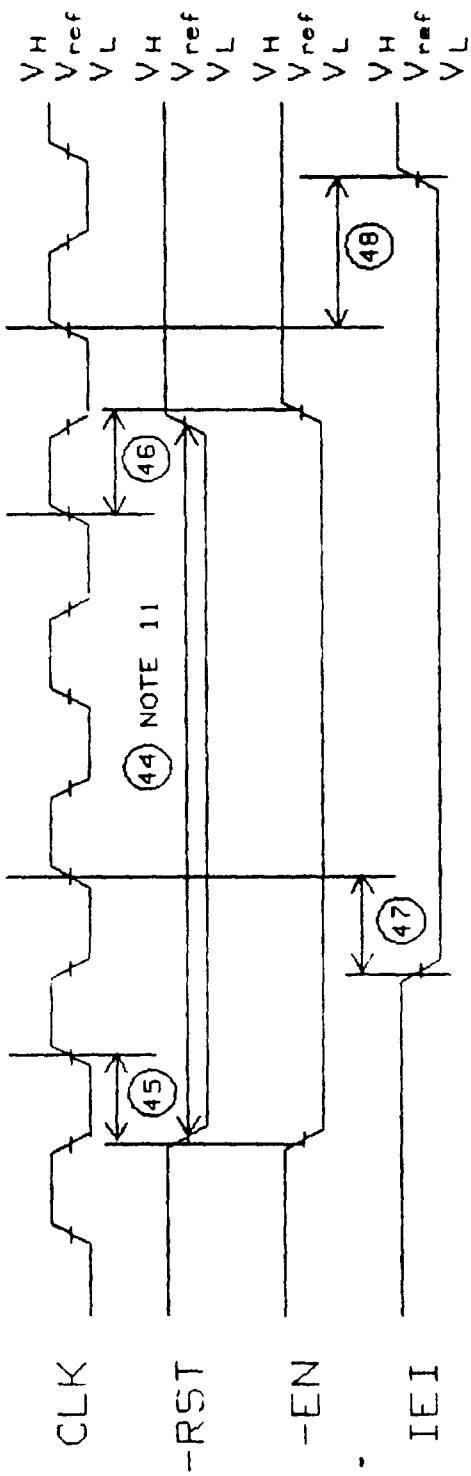


FIGURE 10g. Dynamic waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	4.2	7.0	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 140 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1	0	400	mV	1-3

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	1.044 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D07, inspect for the most negative spike as shown on figure 11.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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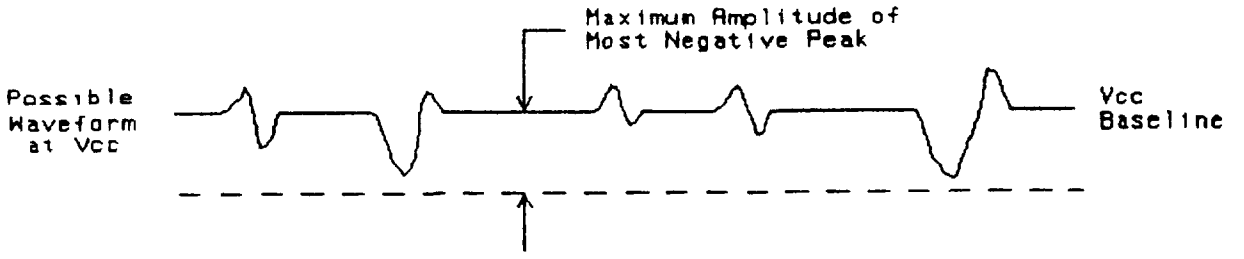


FIGURE 11. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs.
T1	C/T1	14	0	400	mV	1-5	12,13
T2	C/T2	11					
T3	C/T3	22					
T4	PA0	61					
T5	PA1	62					
T6	PA2	13					
T7	PA3	64					
T8	PA4	65					
T9	PA5	16					
T10	PA6	19					
T11	PA7	20					
T12	IEO	21					
T13	Q0	91					
T14	Q1	45					
T15	Q2	95					
T16	Q3	46					
T17	Q4	94					
T18	Q5	96					
T19	Q6	97					
T20	Q7	98	0	400	mV	1-5	12,13

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NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.05 V	T1-T20
Power supply current limit	1.044 A	NA	T1-T20
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T20 T1-T20
Output crosstalk measurement	NA	± 20 mV	T1-T20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 12.

(e) The maximum probe capacitance is 10 pF.

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NOTES: - Continued.

3. The test circuit shall be as indicated on figure 13. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

4. All outputs shall be loaded as specified on figure 7.

5. Abbreviations used are defined in 6.1.

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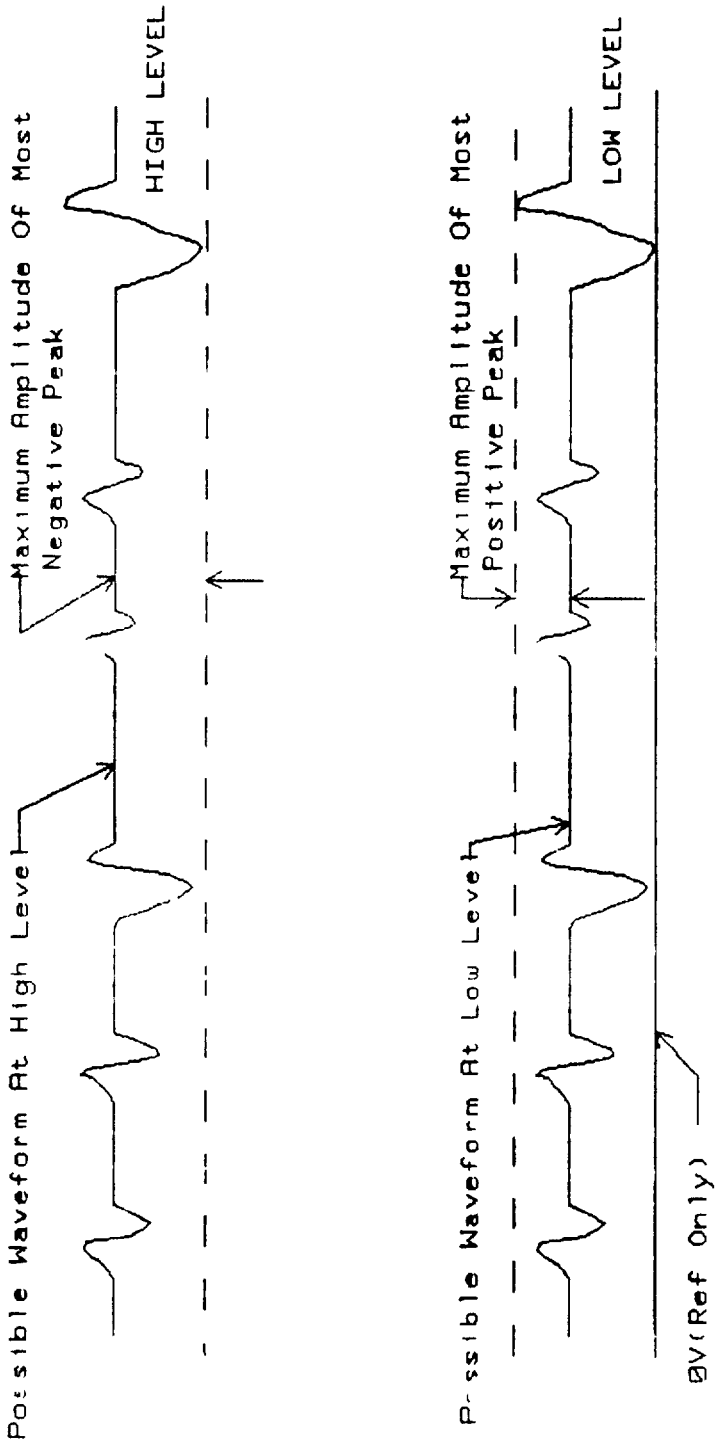


FIGURE 12. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

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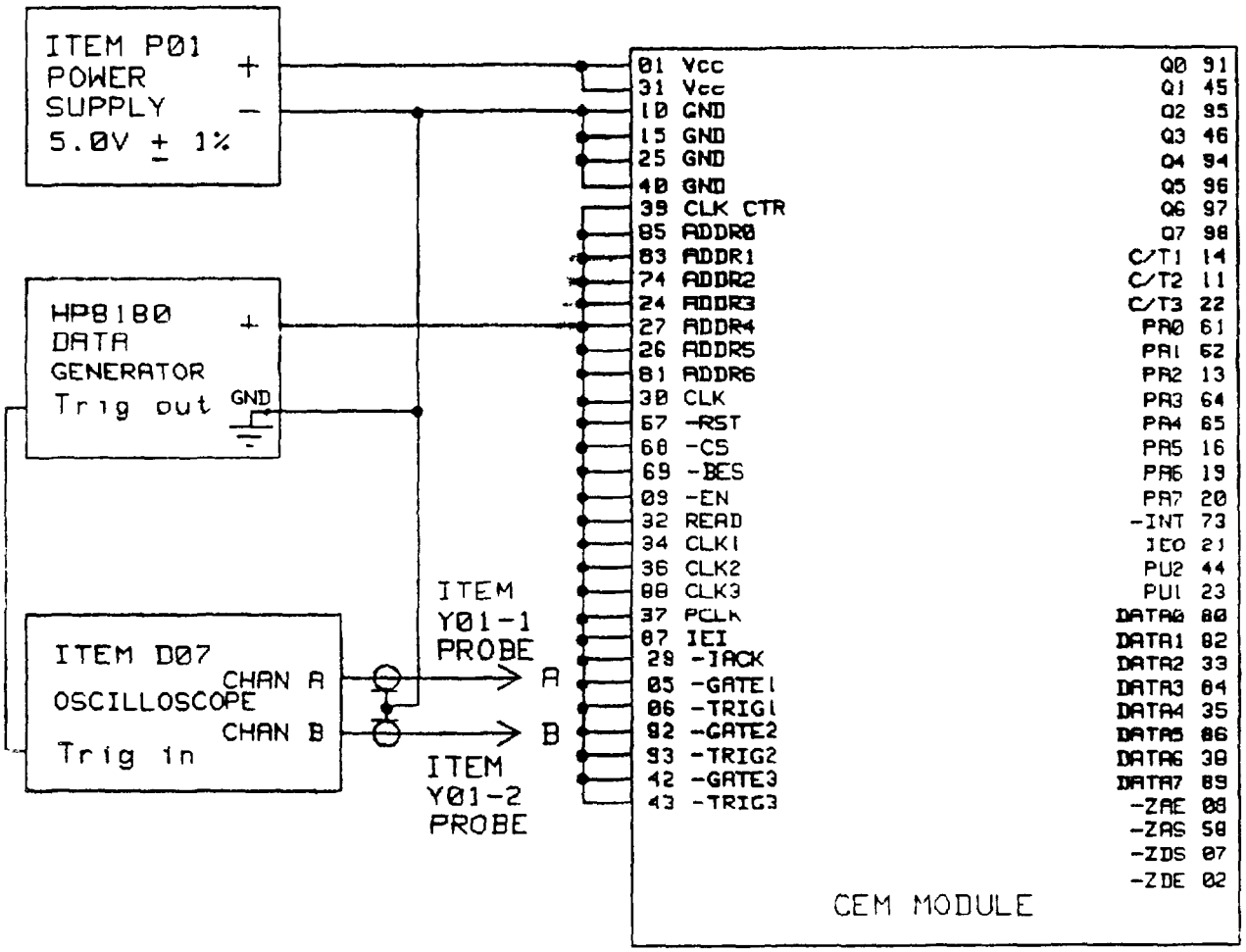


FIGURE 13. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes	Figs.
U1	C/T1	14	0	400	mV	1-5	12,13
U2	C/T2	11					
U3	C/T3	22					
U4	PA0	61					
U5	PA1	62					
U6	PA2	13					
U7	PA3	64					
U8	PA4	65					
U9	PA5	16					
U10	PA6	19					
U11	PA7	20					
U12	-INT	73					
U13	Q0	91					
U14	Q1	45					
U15	Q2	95					
U16	Q3	46					
U17	Q4	94					
U18	Q5	96					
U19	Q6	97					
U20	Q7	98	0	400	mV	1-5	12,13

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NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.05 V	U1-U20
Power supply current limit	1.044 A	NA	U1-U20
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U20 U1-U20
Output crosstalk measurement	NA	± 20 mV	U1-U20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 12.

(e) The maximum probe capacitance is 10 pF.

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NOTES: - Continued.

3. The test circuit shall be as indicated on figure 13. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

4. All outputs shall be loaded as specified on figure 7.

5. Abbreviations used are defined in 6.1.

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TABLE XXII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
V1	READ	32	2	25	pF	1-3
V2	CLK1	34				
V3	CLK2	36				
V4	CLK3	88				
V5	PCLK	37				
V6	IEI	87				
V7	-IACK	29	2	25	pF	1-3
V8	-GATE1	05	2	15	pF	1-3
V9	-GATE2	92				
V10	-GATE3	42				
V11	-TRIG1	06				
V12	-TRIG2	93				
V13	-TRIG3	43				
V14	-RST	67				
V15	-CS	68				
V16	-BES	69	2	15	pF	1-3
V17	ADDR0	85	2	25	pF	1-3
V18	ADDR1	83				
V19	ADDR2	74				
V20	ADDR3	24	2	25	pF	1-3

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TABLE XXII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
V21	ADDR4	27	2	25	pF	1-3
V22	ADDR5	26				
V23	ADDR6	81				
V24	CLK CTR	39	2	25	pF	1-3
V25	CLK	30	2	50	pF	1-3
V26	-EN	09	4	70	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	V1-V26
Capacitance meter voltage	15 mV	± 1.5 mV	V1-V26
Capacitance measurement	NA NA NA NA NA	± 1.15 pF ± 0.65 pF ± 1.15 pF ± 2.4 pF ± 3.3 pF	V1-V7 V8-V16 V17-V24 V25 V26

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIII. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W1	C/T1	14	2	25	pF	1-3
W2	C/T2	11				
W3	C/T3	22	2	25	pF	1-3
W4	-INT	73	2	40	pF	1-3
W5	IEO	21	2	40	pF	1-3
W6	Q0	91	2	20	pF	1-3
W7	Q1	45				
W8	Q2	95				
W9	Q3	46				
W10	Q4	94				
W11	Q5	96				
W12	Q6	97				
W13	Q7	98	2	20	pF	1-3
W14	PA0	61	2	30	pF	1-3
W15	PA1	62				
W16	PA2	13				
W17	PA3	64				
W18	PA4	65				
W19	PA5	16				
W20	PA6	19	2	30	pF	1-3

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TABLE XXIII. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W21	PA7	20	2	30	pF	1-3
W22	DATA0	80				
W23	DATA1	82				
W24	DATA2	33				
W25	DATA3	84				
W26	DATA4	35				
W27	DATA5	86				
W28	DATA6	38				
W29	DATA7	89	2	30	pF	1-3
W30	-ZDE	02	2	60	pF	1-3
W31	-ZDS	07	2	60	pF	1-3
W32	-ZAS	58	2	50	pF	1-3
W33	-ZAE	08	4	85	pF	1-3

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NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W33
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W33
Capacitance measurement	NA	± 1.15 pF	W1-W3
	NA	± 1.9 pF	W4,W5
	NA	± 0.9 pF	W6-W13
	NA	± 1.4 pF	W14-W29
	NA	± 2.9 pF	W30,W31
	NA	± 2.4 pF	W32
	NA	± 4.05 pF	W33

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Threshold test.

Test cond.	Test description	Notes	Figure
X	Threshold test	1-3	8

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	X
Power supply current limit	1.044 A	NA	X
Pattern voltage high, (Voh) low, (Vol)	 2.0 V 0.7 V	 ± 20 mV ± 7 mV	 X X
Output voltage measurement high (Voh) low (Vol)	 NA NA	 ± 105 mV ± 25 mV	 X X

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2. The following procedure is required to perform this test.

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results with the output low and high levels specified in the output tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Abbreviations used are defined in 6.1.

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TABLE XXV. Disable time test

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
High to high impedance (tphz)								
Y1	-EN TO -ZAE	29,32, 67,68, 69	Open	8	0	42	nS	14,17
Y2	-EN TO -ZAS			58				
Y3	-EN TO -ZDS			7				
Y4	-EN TO -ZDE	29,32, 67,68, 69	Open	2	0	42	nS	14,17
Low to high impedance (tplz)								
Y5	-EN TO -ZAE	29	Open	8	0	42	nS	15,18
Y6	-EN TO -ZAS			58				
Y7	-EN TO -ZDS	29		7				15,18
Y8	-EN TO -ZDE	29,67	Open	2	0	42	nS	15,19
High to high impedance (tphz)								
Y9	-BES TO D00	67	68	80	0	75	nS	16,20
Y10	-BES TO D01	67	68	82	0	75	nS	16,20

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TABLE XXV. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
Y11	-BES TO D02	67	68	33	0	75	nS	16,20
Y12	-BES TO D03			84				
Y13	-BES TO D04			35				
Y14	-BES TO D05			86				
Y15	-BES TO D06			38				
Y16	-BES TO D07	67	68	89	0	75	nS	16,20
Low to high impedance (tplz)								
Y17	-BES TO D00	67	68	80	0	75	nS	16,20
Y18	-BES TO D01			82				
Y19	-BES TO D02			33				
Y20	-BES TO D03			84				
Y21	-BES TO D04			35				
Y22	-BES TO D05			86				
Y23	-BES TO D06			38				
Y24	-BES TO D07	67	68	89	0	75	nS	16,20

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NOTES:

1. Disable time test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply	5.0 V	± 50 mV	Y1-Y24
Power supply current	1.044 A	NA	Y1-Y24
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 Y1-Y24 Y1-Y24
Disable time measurement	NA NA	± 2.1 nS ± 3.75 nS	Y1-Y8 Y9-Y24

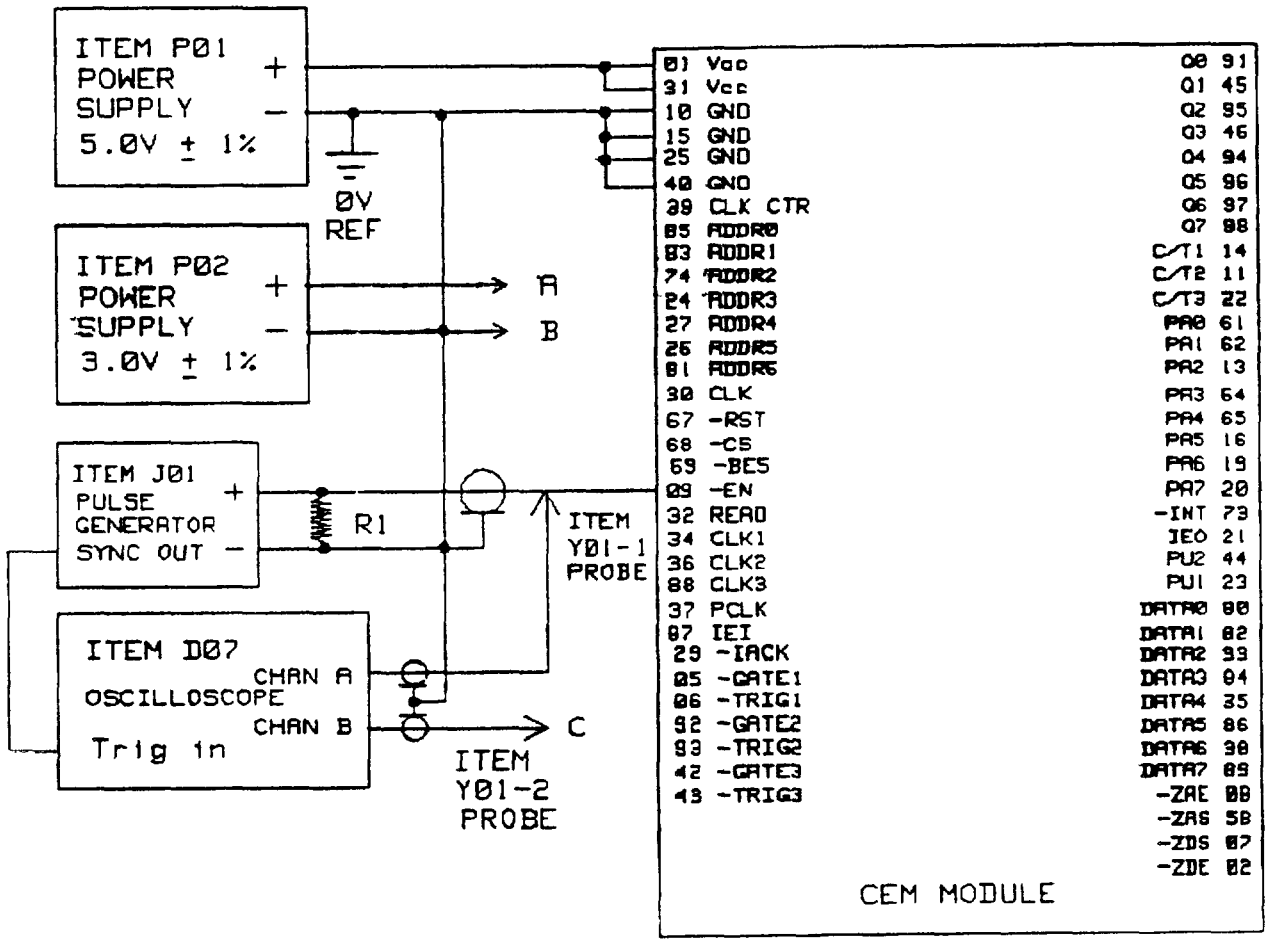
2. The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

3. The test circuit shall be as indicated on figures 14, 15 and 16.

4. Module inputs shall meet the timing requirements shown on figures 17 through 20.

5. Abbreviations used are defined in 6.1.

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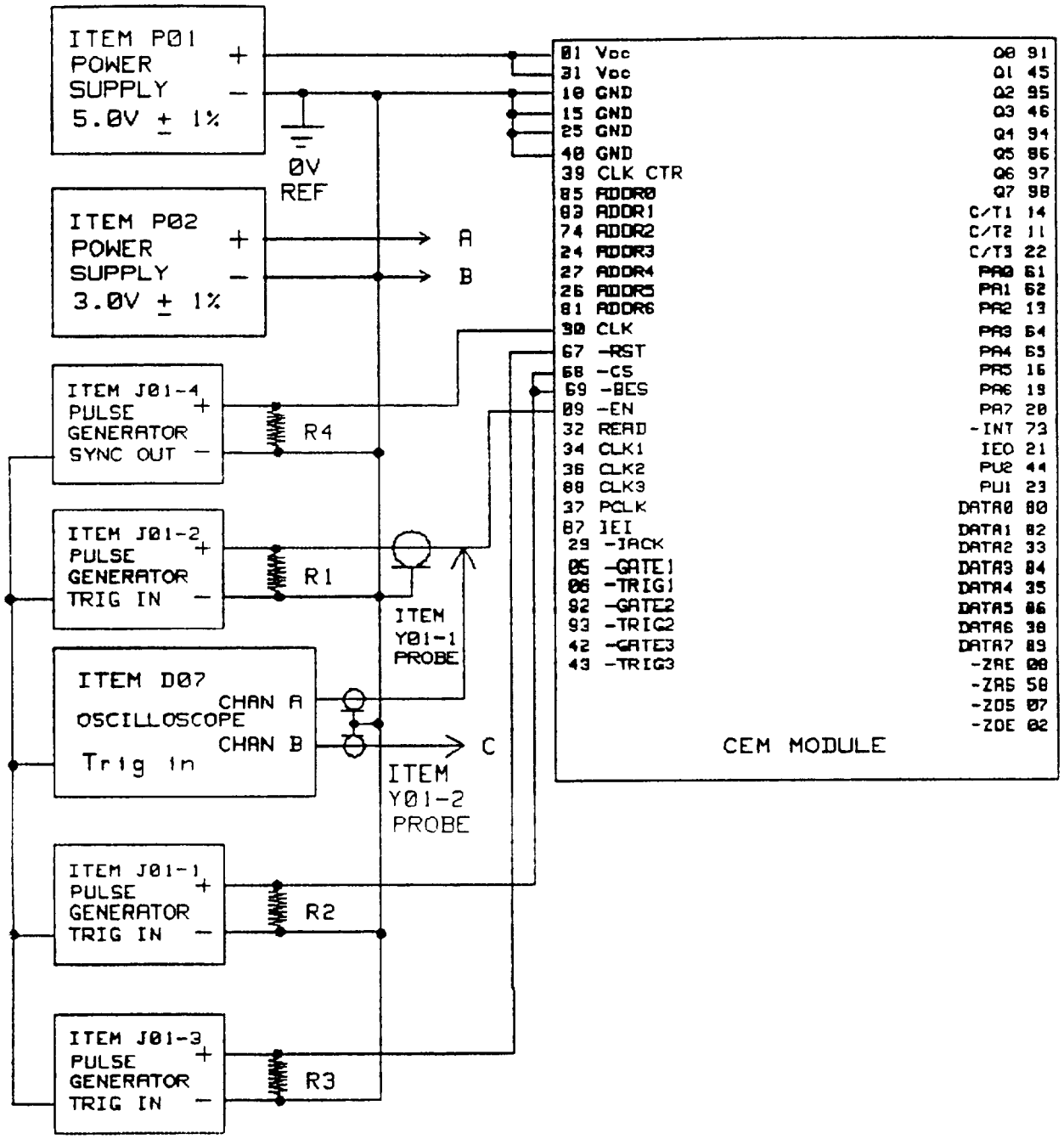


NOTE:

1. R1 = 51 Ohms ± 5 percent, 1/2 W.

FIGURE 14. Disable time test circuit for table XXV.

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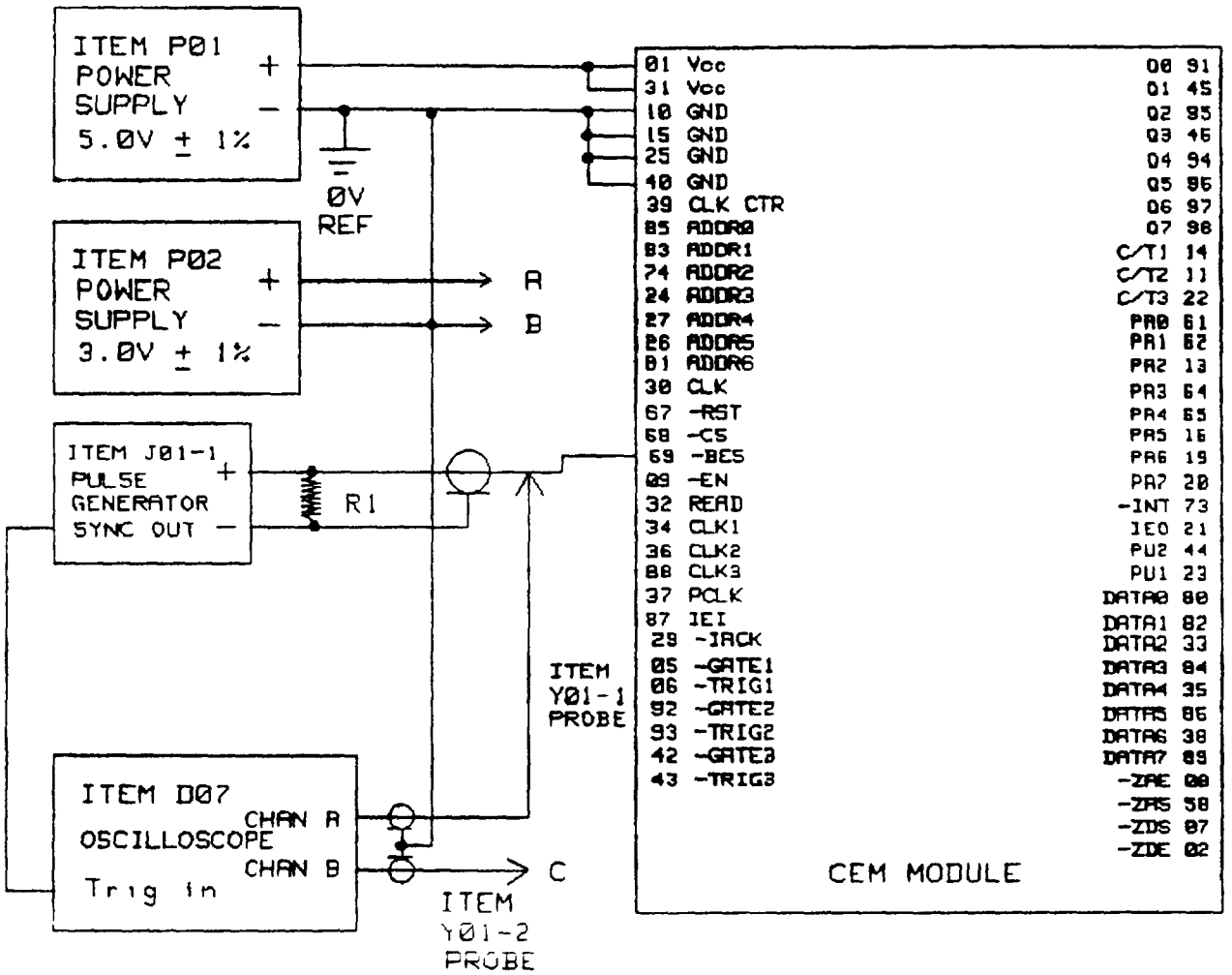


NOTE:

1. R1 = R2 = R3 = R4 = 51 Ohms ± 5 percent, 1/2 W.

FIGURE 15. Disable time test circuit for table XXV.

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NOTE:

1. R1 = 51 Ohms ± 5 percent, 1/2 W.

FIGURE 16. Disable time test circuit for table XXV.

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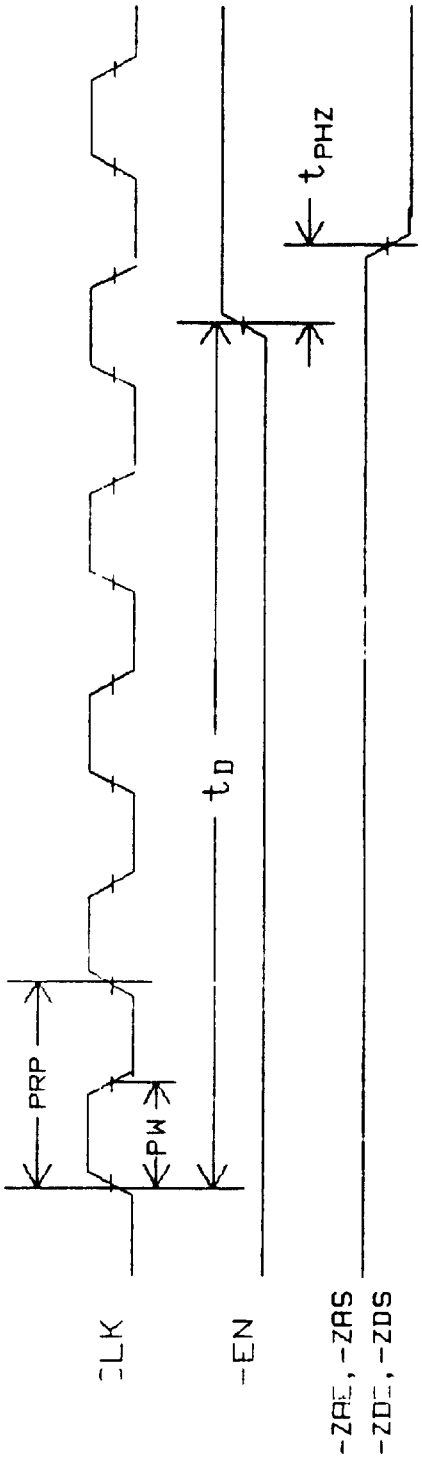


FIGURE 17. Disable time test waveforms for table XXV.

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NOTES:

1. PRP = 200 nS ± 20 nS.
   PW = 100 nS ± 10 nS.

2. tD = 900 nS ± 90 nS.

3. Transition times shall be as specified on figure 9.

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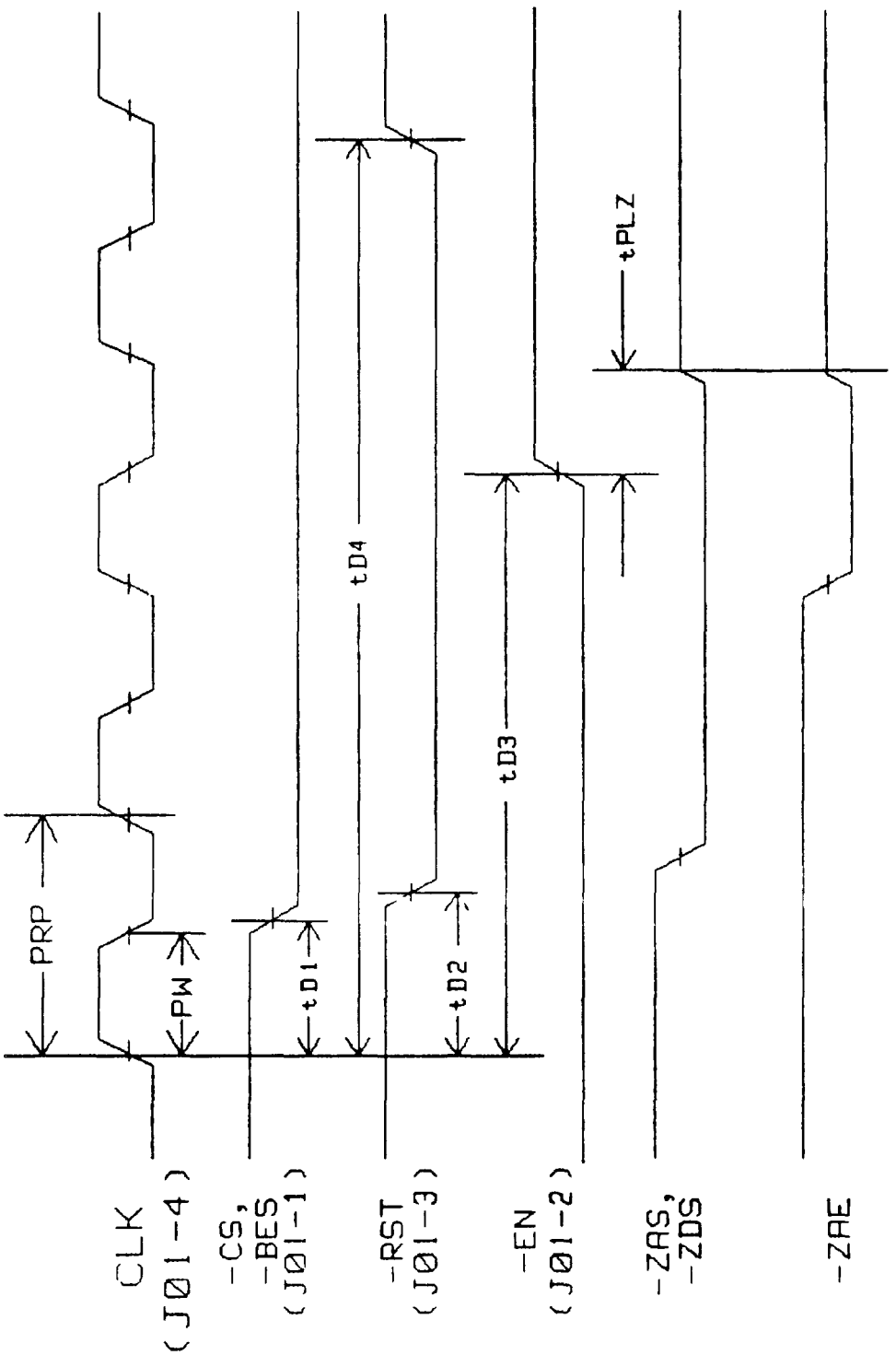


FIGURE 18. Disable time test waveforms for table XXV.

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NOTES:

1. PRP = 200 nS ± 20 nS.
   PW = 100 nS ± 10 nS.

2. tD1 = 100 nS ± 10 nS.
   tD2 = 110 nS ± 10 nS.
   tD3 = 500 nS ± 50 nS.
   tD4 = 700 nS ± 70 nS.

3. Transition times shall be as specified on figure 9.

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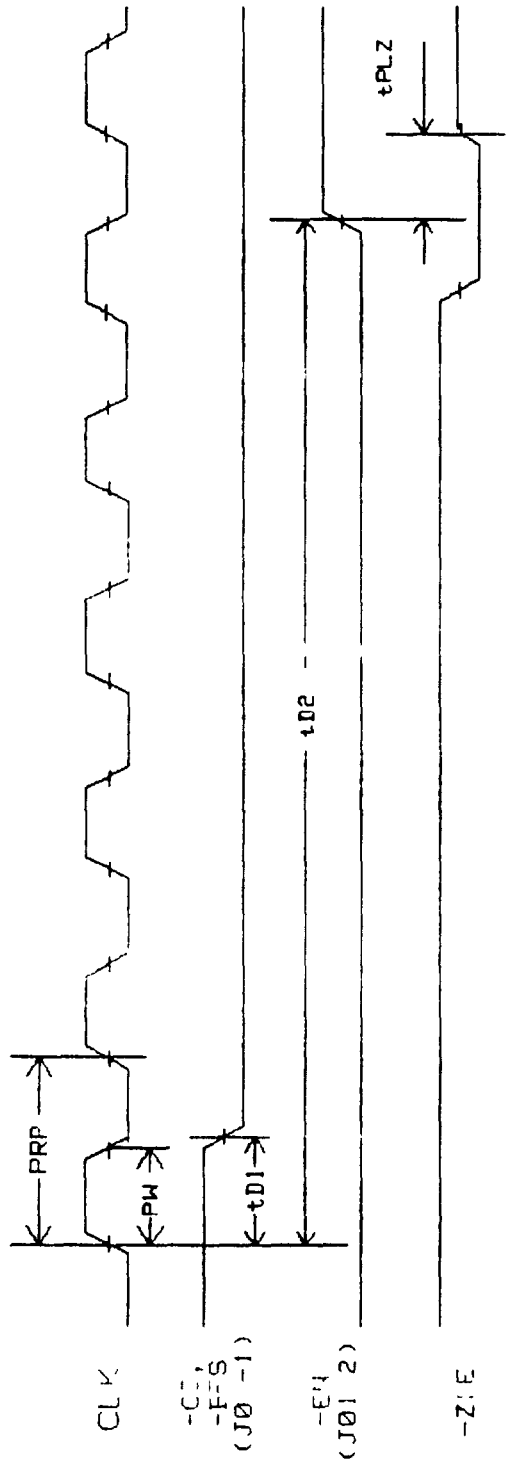


FIGURE 19. Disable time test waveforms for table XXV.

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NOTES:

1. PRP = 200 nS ± 20 nS.
   PW = 100 nS ± 10 nS.

2. tD1 = 100 nS ± 10 nS.
   tD2 = 1100 nS ± 100 nS.

3. Transition time shall be as specified on figure 9.

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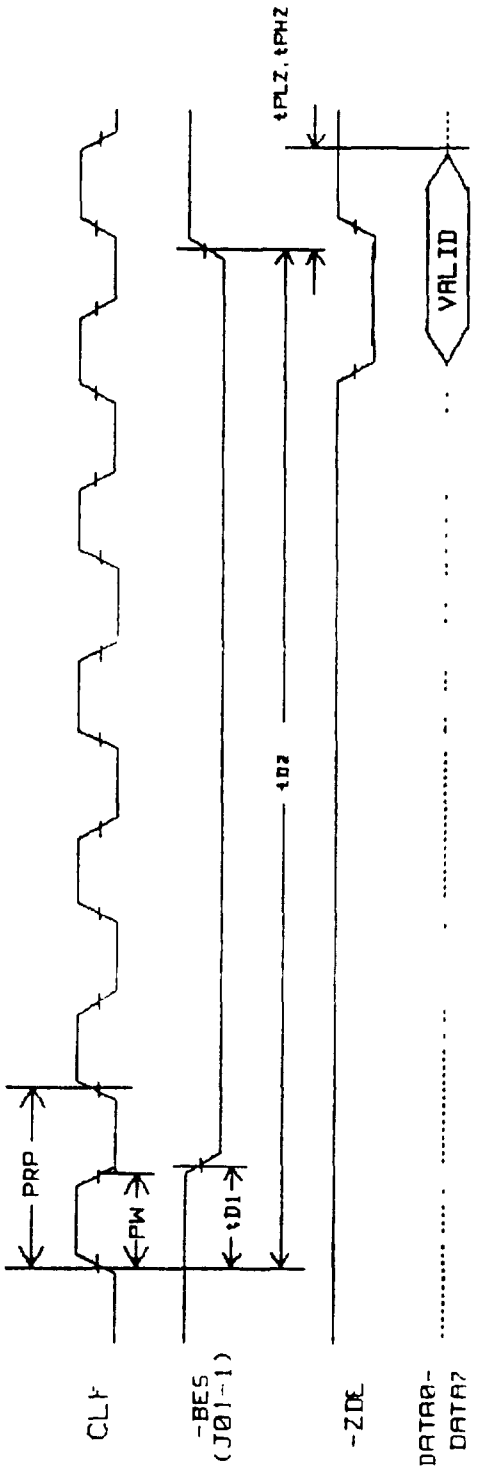


FIGURE 20. Disable time test waveforms for table XXV.

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NOTES:

1. PRP = 200 nS ± 20 nS.
   PW = 100 nS ± 10 nS.

2. tD1 = 100 nS ± 10 nS.
   tD2 = 1100 nS ± 100 nS.

3. Transition times shall be as specified on figure 9.

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APPENDIX

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

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APPENDIX

30. Module function. The CEM module is a general purpose timing controller module. Its primary function is to provide a microprocessor based system with controlled timing functions of various programmable frequencies. The CEM contains a Z8036 Counter/Timer and Parallel I/O (CIO) Unit (see figures 1a-1f of MIL-M-28787/363). The Z8036 contains three 16-bit counters that are software programmable. Interface circuitry includes bus control and buffering logic. Also included on this module is divide logic which accepts a Clock input (CLK CTR) and provides 8 clock outputs, each of which has a frequency less, by a factor of two, than that of the previous output. The general purpose Z8036 bidirectional Port A (PA0-PA7) is pinned out on the module. This port can only be used as an output port since the Z8036 inputs have no protection against NMOS undershoot. The dedicated input bits of general purpose Z8036 Port B (PB1-PB3, PB5-PB7) are controlled by module inputs CLK2,-TRIG2,-GATE2,CLK1,-TRIG1, and -GATE1. The remaining bits (PB0,PB4) are output bits which drive module outputs C/T2,C/T1. (See table I) Similarly, general purpose Port C has three dedicated input bits (PC1-PC3) controlled by module inputs CLK3,-GATE3,-TRIG3 and one dedicated output bit (PC0) which drives module output C/T3. For ease of testing, several Z8036 inputs are brought out to module pins (test points). These are the Z8036 Chip Select (-ZAE), Address Strobe (-ZAS) and Data Strobe (-ZDS) inputs. The LS245 data enable (-ZDE) is also brought out to a module pin. These inputs are to be used as test points only and are not intended for use as general purpose inputs. The automatic tester that is used to drive the -ZAE, -ZAS and -ZDS inputs must have a series damping resistor in its output drive lines to limit input undershoot at the Z8036. Before driving these lines, the module pin -EN must be brought high in order to place the PAL outputs which normally drive the -ZAE,-ZAS and -ZDS signals in the high impedance state. Programming the CEM entails loading the Z8036 control registers with bits to implement the desired operation. Individual enable bits are provided for various major blocks so that erroneous operations do not occur while the Z8036 is being initialized. The initialization sequence that should be followed for the CEM is as follows: First the Z8036 should be reset. The CEM can be reset in either of two ways. The Z8036 CIO is reset by an active (low) -RST input to the CEM. This causes the Bus Control Logic (PAL16R4) to assert Address Strobe (-ZAS), and Data Strobe (-ZDS) internally within the module, causing the Z8036 to be reset. When a 1 is written into bit 0 (Reset bit) of the Master Interrupt Control Register (MICR,figure 3), the Z8036 is also reset. Once reset, the only thing that can be done is to read and write the Reset bit. Writes to all other bits are ignored and all reads return 0's. In this state, all Z8036

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control bits are forced to 0. Only after clearing the Reset bit (by writing a 0 to it) can the other Z8036 registers be configured for module operation. The next step is to configure the remaining bits in the MICR. At a minimum, bit 1 (RJA) must be specified to determine register addressing scheme as shown in table II. The next step is to configure the Counter/Timer Mode Specification Register (figure 7) of the desired Counter/Timer. Initialization requires several things to be specified; for example, the external lines to be used, the output duty cycle and whether the cycle is continuous or single-cycle.

TABLE I. Counter/timer external control inputs/outputs.

Counter /timer	Signal source/ destination	Counter input	Trigger input	Gate input	Counter/ timer output
C/T1	CEM	CLK1	-TRIG1	-GATE1	C/T1
	Z8036	PB5	PB6	PB7	PB4
C/T2	CEM	CLK2	-TRIG2	-GATE2	C/T2
	Z8036	PB1	PB2	PB3	PB0
C/T3	CEM	CLK3	-TRIG3	-GATE3	C/T3
	Z8036	PC1	PC2	PC3	PC0

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TABLE II. Summary of register addresses.

Address						RJA bit = 1 RJA bit = 0	General	Port			C/T		
A5 A6	A4 A5	A3 A4	A2 A3	A1 A2	A0 A1			A	B	C	1	2	3
0	0	0	0	0	0	Master Interrupt Control	X						
0	0	0	0	0	1	Master Configuration Control	X						
0	0	0	0	1	0	Interrupt Vector		X					
0	0	0	0	1	1	Interrupt Vector			X				
0	0	0	1	0	0	Interrupt Vector					X	X	X
0	0	0	1	0	1	Data Path Polarity				X			
0	0	0	1	1	0	Data Direction				X			
0	0	0	1	1	1	Special I/O Control				X			
0	0	1	0	0	0	Port Command and Status		X					
0	0	1	0	0	1	Port Command and Status			X				
0	0	1	0	1	0	Counter Timer Command and Status					X		
0	0	1	0	1	1	Counter Timer Command and Status						X	
0	0	1	1	0	0	Counter Timer Command and Status							X
0	0	1	1	0	1	Port Data		X					
0	0	1	1	1	0	Port Data			X				
0	0	1	1	1	1	Port Data				X			
0	1	0	0	0	0	Counter Timer Current Count MSB					X		
0	1	0	0	0	1	Counter Timer Current Count LSB					X		
0	1	0	0	1	0	Counter Timer Current Count MSB						X	
0	1	0	0	1	1	Counter Timer Current Count LSB						X	

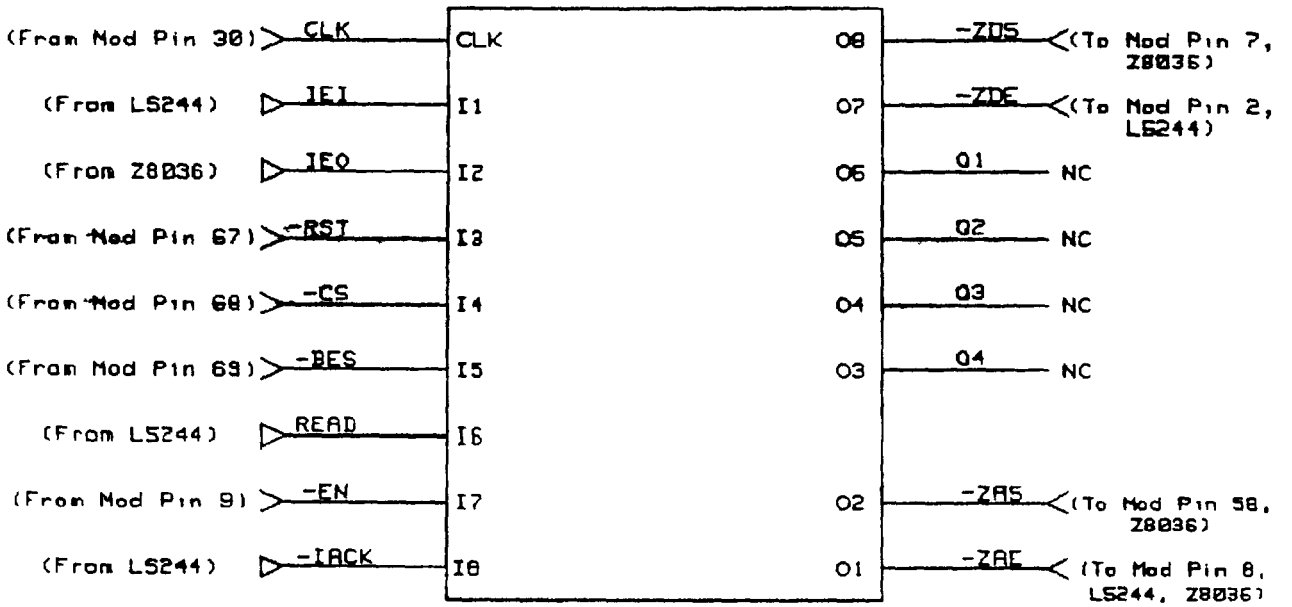
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TABLE II. Summary of register addresses - Continued.

Address						RJA bit = 1 / RJA bit = 0	General	Port			C/T		
A5 / A6	A4 / A5	A3 / A4	A2 / A3	A1 / A2	A0 / A1			A	B	C	1	2	3
0	1	0	1	0	0	Counter Timer Current Count MSB							X
0	1	0	1	0	1	Counter Timer Current Count LSB							X
0	1	0	1	1	0	Counter Timer Time Constant MSB					X		
0	1	0	1	1	1	Counter Timer Time Constant LSB					X		
0	1	1	0	0	0	Counter Timer Time Constant MSB						X	
0	1	1	0	0	1	Counter Timer Time Constant LSB						X	
0	1	1	0	1	0	Counter Timer Time Constant MSB							X
0	1	1	0	1	1	Counter Timer Time Constant LSB							X
0	1	1	1	0	0	Counter Timer Mode Specification					X		
0	1	1	1	0	1	Counter Timer Mode Specification						X	
0	1	1	1	1	0	Counter Timer Mode Specification							X
0	1	1	1	1	1	Current Vector	X						
1	0	0	0	0	0	Port Mode Specification		X					
1	0	0	0	0	1	Port Handshake		X					
1	0	0	0	1	0	Data Path Polarity		X					
1	0	0	0	1	1	Data Direction		X					
1	0	0	1	0	0	Special I/O Control		X					
1	0	0	1	0	1	Pattern Polarity		X					
1	0	0	1	1	0	Pattern Transition		X					
1	0	0	1	1	1	Pattern Mask		X					
1	0	1	0	0	0	Port Mode Specification			X				
1	0	1	0	0	1	Port Handshake			X				
1	0	0	0	1	0	Data Path Polarity			X				
1	0	1	0	1	1	Data Direction			X				
1	0	1	1	0	0	Special I/O Control			X				
1	0	1	1	0	1	Pattern Polarity			X				
1	0	1	1	1	0	Pattern Transition			X				
1	0	1	1	1	1	Pattern Mask			X				

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APPENDIX



PAL Booleans

$-ZAE = Q2 + \overline{Q4}$

$-ZAS = (Q2 + \overline{Q3}) \cdot -RST$

$-ZDE = (\overline{Q1} + Q4 + -CS) \cdot (-ZDE + -BES) \cdot (\overline{Q1} + Q4 + IEO + \overline{IEI})$

$-ZDS = (\overline{Q2} + Q4 + READ) \cdot (-ZDE + \overline{READ} + \overline{Q2}) \cdot -RST$

$Q1(n+1) = (-BES(n+1) + \overline{Q4}(n) + -ZDE(n+1)) + (-IACK(n+1) \cdot -CS(n+1))$

$Q2(n+1) = Q1(n) + (-CS(n+1) \cdot -IACK(n+1))$

$Q3(n+1) = Q2(n)$

$Q4(n+1) = Q3(n)$

FIGURE 1. PAL16R4 description.

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Next, the Time Constant must be specified by writing the desired value to the Time Constant Register (TCR). The TCR (figure 6) is accessible as two 8-bit registers. The registers are readable as well as writeable, and can be accessed in any order. A 0 in the TCR specifies a Time Constant of 65,536. The Data Path Polarity and Data Direction registers (figure 4) must be setup for each of the ports. Finally, the Counter/Timer Enable and Port Enable bits in the Master Configuration Control Register are set. This allows normal operation to commence. Programmable Array Logic (PAL16R4) is used to implement the necessary timing sequence for the Z8036. It provides separation of -AS and -DS inputs to the Z8036 as well as data and address bus control. A description of the PAL logic is shown on figure 1. In order to maintain the minimum pulse width specifications of the Z8036, the maximum allowable frequency for the Counter/Timer inputs (CLK1-CLK3) is 1.5 MHz. The maximum allowable frequency for PCLK is 3.0MHz. The three independent 16-bit counter/timers each consist of a presettable 16-bit down counter, a 16-bit Time Constant Register, a 16-bit Current Count Register, an 8-bit Mode Specification Register, an 8-bit Command and Status Register, and the associated control logic that links these registers (see figure 1f of M28787/363). There are up to four lines of direct external control provided for each counter/timer (refer to table I). These are the Counter Input, Trigger Input, Gate Input and Counter/Timer Output. It is assumed in the following discussion that all Z8036 bit ports are programmed as non-inverting (see figure 4). The rising edge of the Trigger Input causes the down counter to be loaded with the value stored in the Time Constant Register. The countdown will proceed as long as the Gate Input is high. In the counter mode, the down counter is decremented on the rising edge of the Counter Input. If the Gate input goes low, the countdown halts. It resumes when the Gate Input goes high again. Both the Gate and Trigger inputs can be controlled through software (see C/T Command and Status register description). The rate at which the down counter counts is determined by the mode of the counter/timer (see C/T Mode Specification register description). In the timer mode, the down counter is clocked internally by a signal that is half the frequency of the Peripheral Clock (PCLK) input to the module. Counter/Timers 1 and 2 can be linked internally in three different ways. Counter/Timer 1's output (inverted) can be used as Counter/Timer 2's Trigger, Gate or Counter input. When linked, the Counter/Timers have the same capabilities as when used separately. The only restriction is that when Counter/Timer 1 drives Counter/Timer 2's count input, Counter/Timer 2 must be programmed with its external count input disabled.

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The CEM has four potential sources of interrupts generated by the Z8036 CIO. These are the three counter/timers and Port A. The priority of these sources are fixed in the following order: Counter/Timer 3, Port A, Counter/Timer 2, Counter/Timer 1. The status registers for each source of interrupt contain the control bits needed for interrupt operations. As part of the Interrupt Acknowledge cycle, the Z8036 is capable of responding with an 8-bit interrupt vector that specifies the source of the interrupt. There is one register that is shared by the three counter/timers and one for Port A. The NMOS, bidirectional signals C/T1 thru C/T3 and PA0 thru PA7 are programmed as output only. The input circuitry, although inactive, can still be damaged by negative voltage spikes greater than 500 mV. This damage will cause the output to fail. Users of this module must ensure that voltage spikes of no greater than -500 mV exist on these lines due to crosstalk or other noise related phenomena.

30.1 CEM control registers. The registers in the CEM are used to control counter activity and are programmed by the microprocessor. Table II lists the addresses of the registers in the CEM. The Z8036 features two ways for register addressing. Both ways use only six of the seven bits of the Address bus. The addressing scheme used is determined by the Right Justify Address (RJA) bit in the Master Interrupt Control register (figure 3). When RJA equals 0, Address Bus bit 0 is ignored, and bits 1 through 6 are decoded for the register address. When RJA equals 1, bits 0 through 5 are decoded for the register address. In the following register descriptions, only six bits are shown for addresses and represent Address bus bits 0 through 5 or 1 through 6, depending on the state of the RJA bit. These registers include the Master Configuration Control, Master Interrupt Control, Data Path Polarity, Data Direction, Special I/O Control, Port Mode Specification Registers, Port Command and Status Registers, and Port Data Register. Each of these registers and those discussed before are described in detail in the following paragraphs. Other registers in the Z8036 cannot be used with the module configuration.

30.2 Master configuration control register. The Master Configuration Control register contains the control bits used to enable different sections of the Z8036 during initial configuration, as well as the bits used to link the ports together and the timers together. All bits are cleared to 0 by resetting the device. This register is read/write, and is illustrated on figure 2. Bit 7, (PBE) when set to a 1, allows Port B to operate normally. When cleared to a 0, it inhibits the Port B logic from issuing an interrupt request.

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The purpose of this bit is to allow Port B to be configured initially without erroneously generating an interrupt or having its I/O lines go low-impedance until it is safe to do so. When bit 6 (CT1E) is cleared to 0, Counter/Timer 1 is put into an initialized state. Setting this bit to a 1 allows the counter/timer to function normally. Bit 5 (CT2E) performs the same function for Counter/Timer 2 that bit 6 performs for Counter/Timer 1. Bit 4 (PCE and CT3E) performs the same function for Counter/Timer 3 and Port C as bits 6 and 7 do for Counter/Timer 1 and Port B, respectively. The CEM module design does not allow the full capability associated with Bit 3 (PLC). All writes to Bit 3 should be 0. Bit 2 (PAE) performs the same function for Port A that bit 7 performs for Port B. Bits 0 and 1 (LC) specify how and if Counter/Timers 1 and 2 are linked.

30.3 Master interrupt control register. The Master Interrupt Control register (see figure 3) contains the primary control bits for the interrupt control logic. The RJA bit (D1) defines the Z8036 addressing scheme. When the module is reset, all bits are forced to 0 except RESET (bit0), which is set to 1. All bits in the Master Interrupt Control register are read/write. Clearing bit 7 (MIE) to a 0 inhibits the device from requesting an interrupt or responding to an Interrupt Acknowledge. A 1 in this bit allows the interrupt logic to operate normally. When bit 7 is set to a 1, reading the base vector register always includes status, independent of the state of bit 4 (2). When bit 7 is set to a 0, reading the base vector register returns the unmodified base vector so that it can be verified. If bit 6 (DLC) is set to 1, the Interrupt Enable Output (IEO) of the module is forced low, disabling interrupts from all lower-priority devices when more than one CEM is used in an interrupt driven mode. When the bit is 0, IEO operates normally. When bit 5 (NV) is set to a 1, the device is inhibited from outputting an interrupt vector during an Interrupt Acknowledge cycle. This allows the vector to be provided by external hardware. If the bit is written with a 0, the interrupt vector is output as usual. If bit 4 (PA VIS) is a 0 when a Port A interrupt is acknowledged, the interrupt vector that is output is the unmodified content of the Port A Interrupt Vector register. If this bit is written with a 1, the Port A base vector is modified to include status, which indicates the cause of the interrupt. The capability associated with bit 3 (PB VIS) is not supported in the CEM module design because Port B is dedicated to counter/timer functions. All writes to bit 3 should be 0. Bit 2 (CT VIS) operates the same way as bit 4 and controls the base interrupt vector shared by the three counter/timers.

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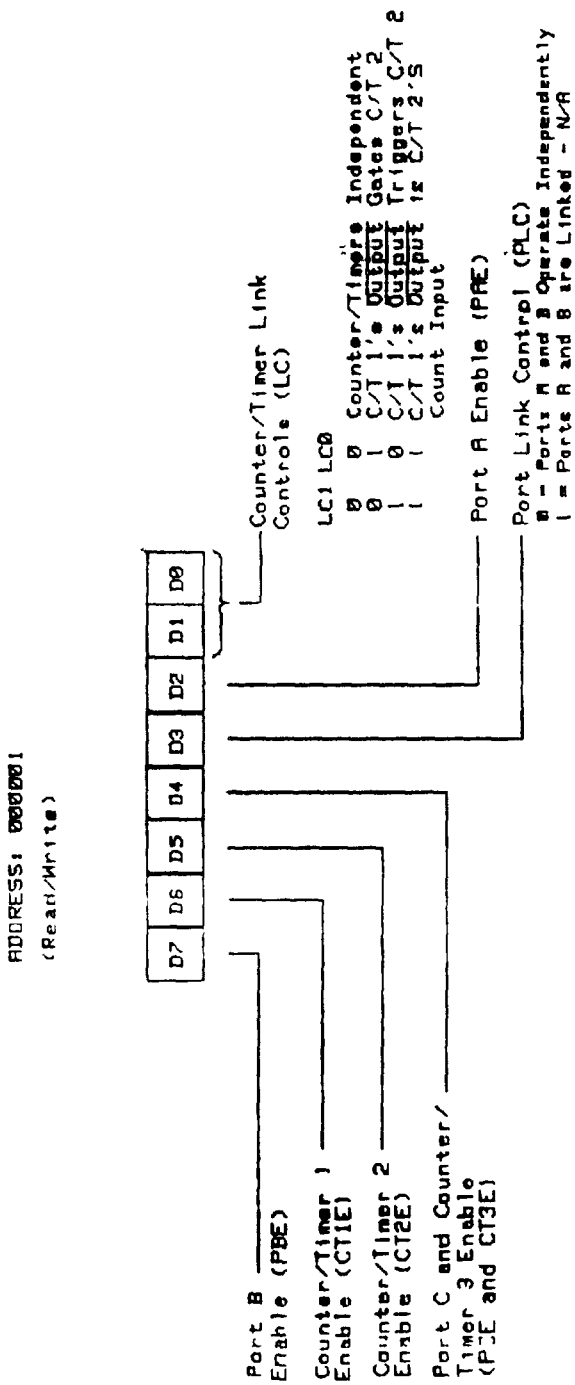
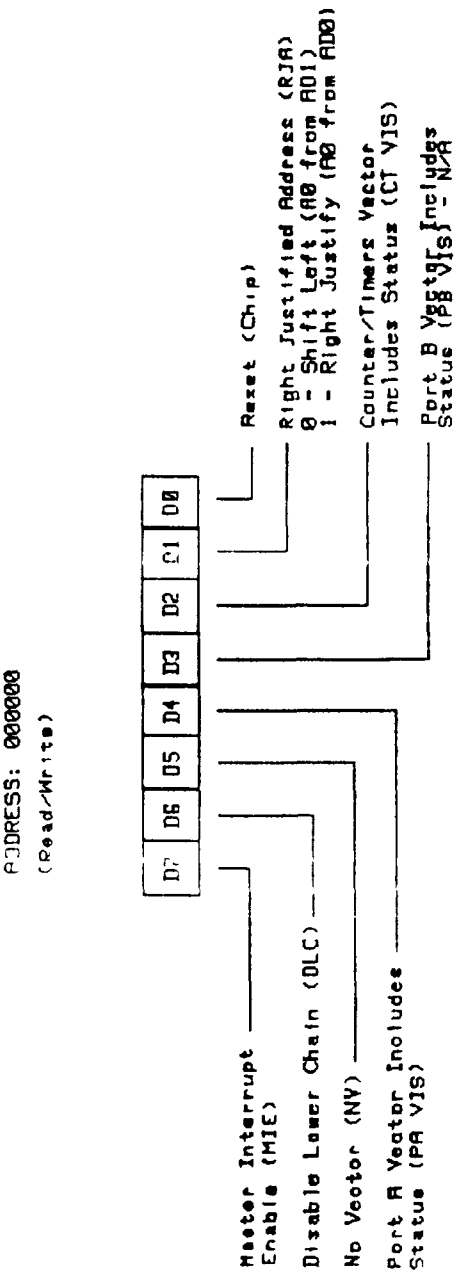


FIGURE 2. Master configuration control register.

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NA = Not available in CEM design.

FIGURE 3. Master interrupt control register.

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30.4 Data path polarity register. The 8-bit data path polarity registers determine whether the data shall be provided non-inverted or inverted by setting each bit (4 LSBs for Port C). A 0 provides a non-inverting path. A 1 will invert all data at the bit position set (see figure 4). A reset clears all bits to 0.

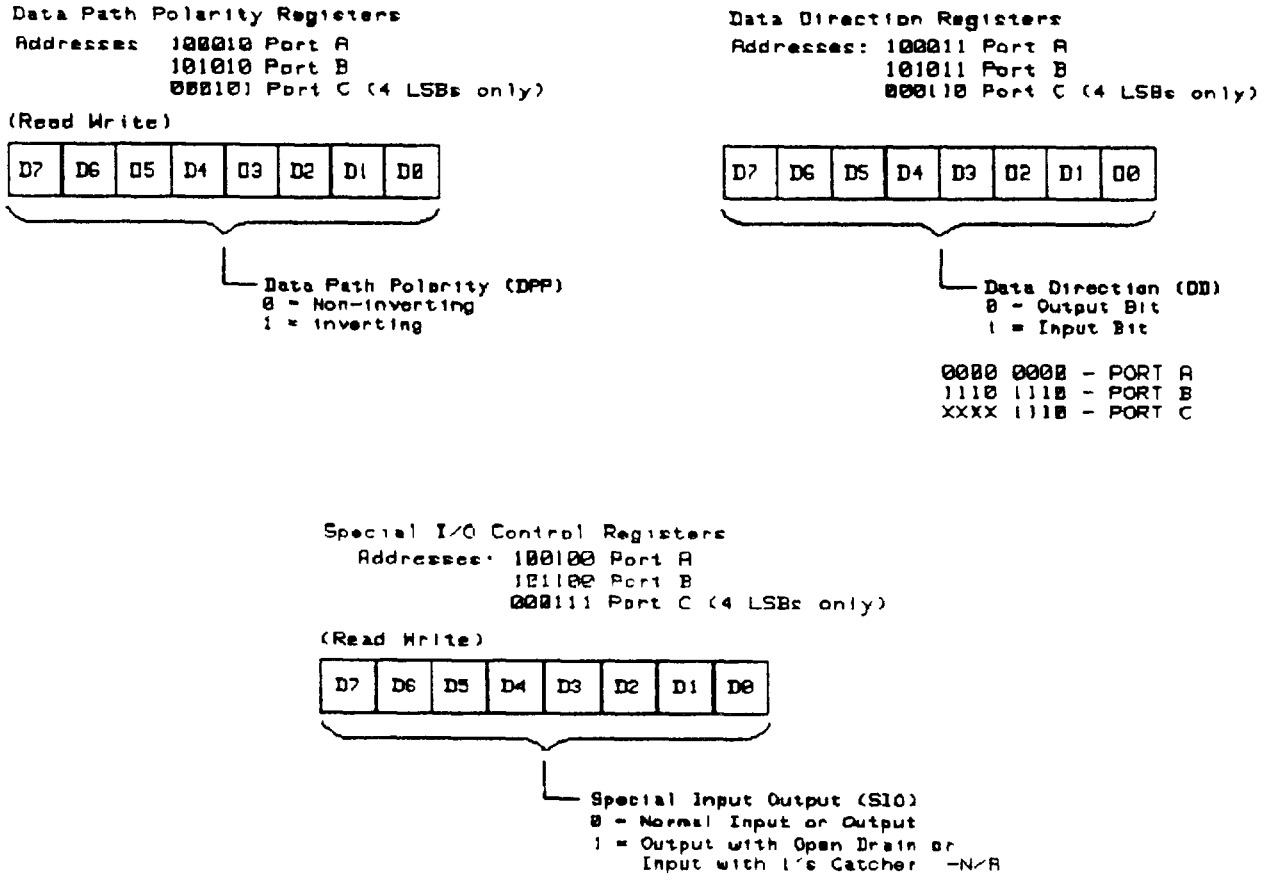
30.5 Data direction register. The 8-bit data direction registers determine whether each bit path will be used for input or output. A 0 provides for an output path. A 1 provides for an input path. Port C uses only the four least significant bits (see figure 4). The CEM module implementation requires that all writes to this register contain the pattern shown on figure 4. A reset forces all bits in these registers to 0.

30.6 Special I/O control register. The 8-bit Special I/O Control Register defines the characteristics of the port input/output bits. For Ports A, B, and C, all bits are set to 0 thus making the bits normal outputs (see figure 4). A reset forces all bits to 0.

30.7 C/T command and status register. The 8-bit Counter/Timer Command and Status Register provides a software gate and trigger input to the counter/timer. It also provides count-in progress status and interrupt control. A reset forces all bits to 0 (see figure 5). Bit 7 (IUS) is automatically set to 1 if its corresponding Interrupt Pending (IP) is the highest-priority interrupt request pending when an Interrupt Acknowledge sequence takes place. It can also be set and cleared by command from the microprocessor using the code shown on figure 5. Its primary function is to control the interrupt daisy chain. When set, it disables lower priority sources in the daisy chain, so that lower priority interrupt sources do not request service while higher priority devices are being serviced. Bit 6 (IE) enables or disables the counter/timer's interrupt logic. When cleared to a 0, the counter/timer is unable to request an interrupt or respond to an Interrupt Acknowledge. If this bit is programmed to 1, the interrupt logic operates normally. It can be set and cleared by command from the microprocessor using the code shown on figure 5. Bit 5 (IP) is a status bit which, when set to 1, indicates that the counter/timer requires servicing. It is automatically set to 1 each time the counter/timer reaches it terminal count (count of 0). The setting of IP results in forcing the CEM -INT output low, if the associated IE bit is set to a 1. It can also be set and cleared by command from the microprocessor using the code shown on figure 5.

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N/A = Not available in CEM design.

FIGURE 4. Bit path definition registers.

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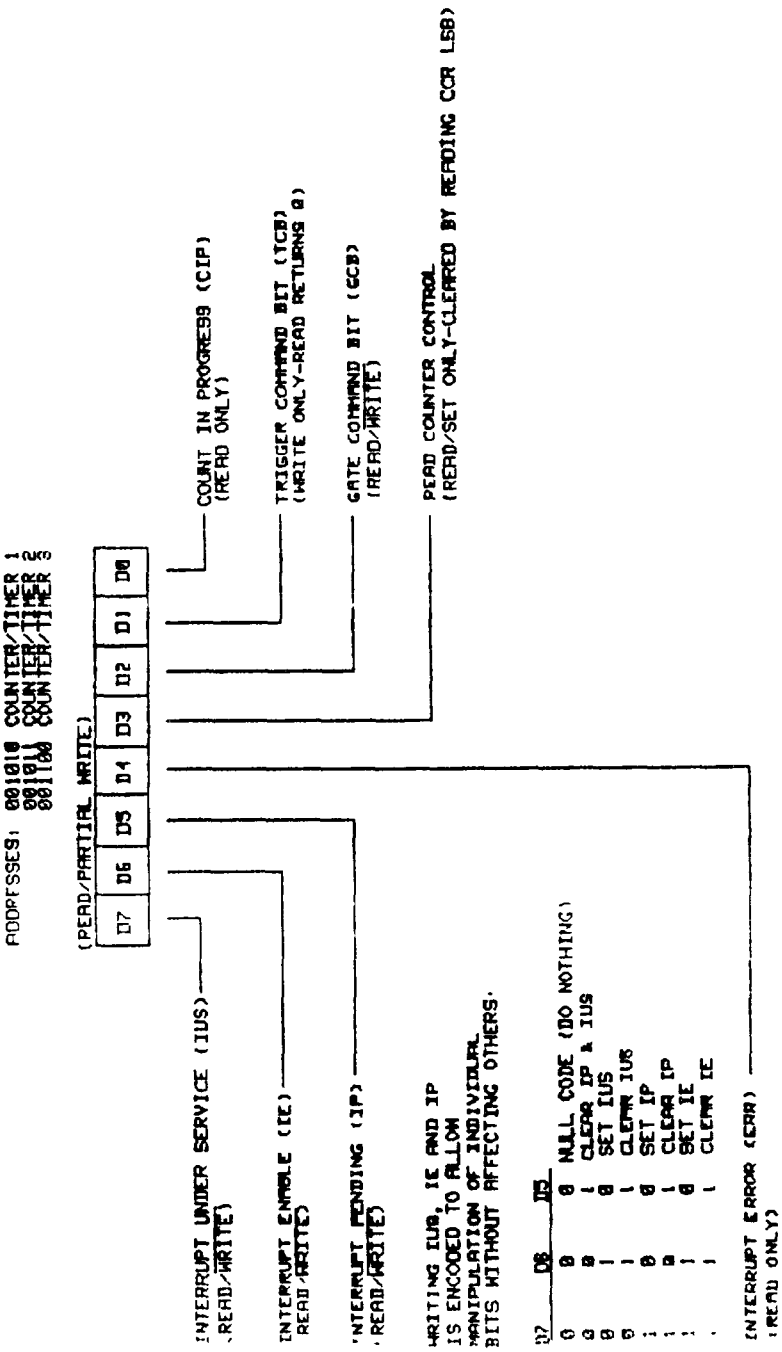


FIGURE 5. Counter/timer command and status register.

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Bit 4 (ERR) is a status bit which is set when an error has occurred. An error occurs for a counter/timer whenever terminal count is reached and IP (bit 5) is still set from a previous terminal count. ERR can be cleared only by having the microprocessor clear the corresponding IP bit. This bit is read only; writes to it are ignored. Bit 3 (RCC) is a command bit that enables the counter/timer to be read reliably while it is in a countdown sequence. Writing a 1 to this bit causes the contents of the Counter/Timer Current Count register (CCR), which normally follows the down counter, to be frozen until the LSB of the CCR is read, at which point the bit is automatically reset. This bit cannot be set unless the counter/timer is enabled. Bit 2 (GCB) is the Gate command bit and is used to start/stop a countdown sequence. By writing a 0 to this bit, the countdown sequence is halted. Returning the bit to 1 allows the sequence to resume where it left off. The state of the GCB bit does not affect the operation of the trigger inputs. Bit 1 (TCB) is the Trigger command bit. Writing a 1 to this bit loads the counter/timer with the contents of the Time Constant Register and causes a countdown sequence to be initiated in the same way as a hardware trigger. It can also retrigger the counter/timer if the Retrigger Enable bit (REB) is set to a 1 in the Counter/Timer Mode Specification register. Bit 0 (CIP) is a status bit that indicates if a countdown sequence is in progress. It is automatically set to 1 when the counter/timer is triggered and the down-counter is loaded with the time constant value. It is automatically reset to 0 when the down-counter reaches a count of 0. The state of the gate inputs (internal and external) has no effect on this bit.

30.8 C/T current count register. This 16-bit register contains the current count contained in the down counter. This register can be read anytime. However, reading the register is asynchronous to the counter's counting, and the value returned is valid only if the counter is stopped. The down counter can be reliably read "on the fly" by first writing a 1 to bit 3 in the C/T Command and Status Register. This freezes the value in the Current Count Register until a read of the least significant byte is performed. (see figure 6) A reset forces the current count register to follow the down counter (neither are forced to a specific value).

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30.9 C/T time constant register. This 16-bit register contains the count to be loaded into the down counter by the trigger input (see figure 6). These registers can be read and written at any time. However, care must be taken when writing to them so that a trigger does not occur while the time constant value is changing. A reset does not affect the Time Constant register. A 0 value in the register specifies a time constant of 65,536.

30.10 C/T mode specification register. This 8-bit register controls the configuration of the counter/timer. It enables the external inputs, output modes, and the procedure upon terminal count (see figure 7). A reset forces all bits to 0. A 0 value in the register specifies a time constant of 65,536. If bit 7 (C/SC) is set to 1, then each time the down-counter reaches the count of 1, the time constant value is reloaded (on the next count) and the countdown sequence is repeated. If this bit is set to 0, the count will terminate at 0. By programming bit 6 (EOE) to be 1, the output of the counter/timer is provided on the module output pin (C/T1, C/T2, C/T3) associated with that particular counter/timer. When bit 5 (ECE) is set to 1, the counter/timer is put into counter mode. The associated module input for each counter/timer (CLK1,CLK2,CLK3) is used as an external counter input. When bit 5 is set to a 0, the counter/timer is put into the timer mode and the down counter is clocked internally by a signal that is half the frequency of the PCLK input to the module. When bit 4 (ETE) is set to 1, the module trigger inputs (-TRIG1,-TRIG2,-TRIG3) are used as trigger inputs to their respective counter/timers. A rising edge (when the data path is specified non-inverting) on this line will cause the down counter to be loaded. To guarantee that the counter/timer will be triggered on a particular rising edge of the clocking signal (PCLK/2 or CLK 1,2,3), the trigger rising edge must satisfy a setup time to the preceding falling edge of the clocking signal (see figure 10c of MIL-M-28787/363). When bit 3 (EGE) is set to 1, the module gate inputs (-GATE1,-GATE2,-GATE3) are used as gate inputs to their respective counter/timers. If the external gate input is a 0 (assuming the data path is programmed non-inverting), the countdown sequence is suspended; forcing it to a 1 enables the countdown sequence to continue. To guarantee the enabling or disabling of the counter/timer for a particular rising edge of the clocking signal (PCLK/2 or CLK 1,2,3), the gate input must satisfy a setup time to the preceding falling edge of the clocking signal (see figure 10c of MIL-M-28787/363). If bit 2 (RE) is set to 0, triggers (external or internal) which occur during a countdown sequence are ignored. If the bit is a 1, each

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trigger causes the time constant value to be reloaded and a new countdown sequence to be initiated. When a counter/timer is programmed in square-wave mode, a retrigger will cause the Time Constant value to be reloaded and the new countdown will start on the first half of the square wave cycle. Bits 1 and 0 (DCS) are used to select the output duty cycle according to the information indicated on figures 7 and 8. When the pulse mode is specified, the output goes to a 1 for one clock cycle, beginning when the down counter leaves the count of 1. In the one-shot mode, the output goes to a 1 when the counter/timer is triggered and goes to a 0 when the down counter reaches 0. When the square wave output duty cycle is specified, the counter/timer goes through two full sequences for each cycle. If a 1 count is detected on the down counter's clocking edge, the output goes to a 1 and the time constant value is reloaded. On the clocking edge, when both the down counter and the output are 1s, the output is pulled back to a 0.

30.11 Port mode specification registers. Each Port Mode Specification register contains the bits that define the port operating mode and specify the operation of the port pattern match logic. Only Port A is available as a general purpose output port on the CEM. All ports must be set up as bit ports in order to use them in the CEM module configuration. See figure 9 for bit assignments. A reset forces all bits to 0. Bits 7 through 3 must be set to 0, since all Z8036 ports are only used as bit ports in the CEM. Bits 2 and 1 (PMS) define the mode of operation of the pattern match logic for Port A only. When operating in the AND or OR mode, it is the transistion from a no-match to a match state that causes the interrupt. In the OR mode, if a second match occurs before the first match goes away, it does not cause a second interrupt. Since a match condition may only last a short time when edges are specified, care must be taken to avoid losing a match condition. In the OR-Priority Encoded Vector mode, interrupts are generated as long as any match state exists. A transition from a no-match to a match state is not required. Only transparent mode (LPM = 0) is supported in this mode. Bit 0, (LPM) when set to 1, causes the port to latch the input data present at the port when a pattern match is detected and to hold the data until the IP is cleared (figure 10). If the bit is 0, (transparent mode) pattern matches are still detected, but the data read back from the port follows the port pins and does not necessarily indicate the state of the port at the time of the match.

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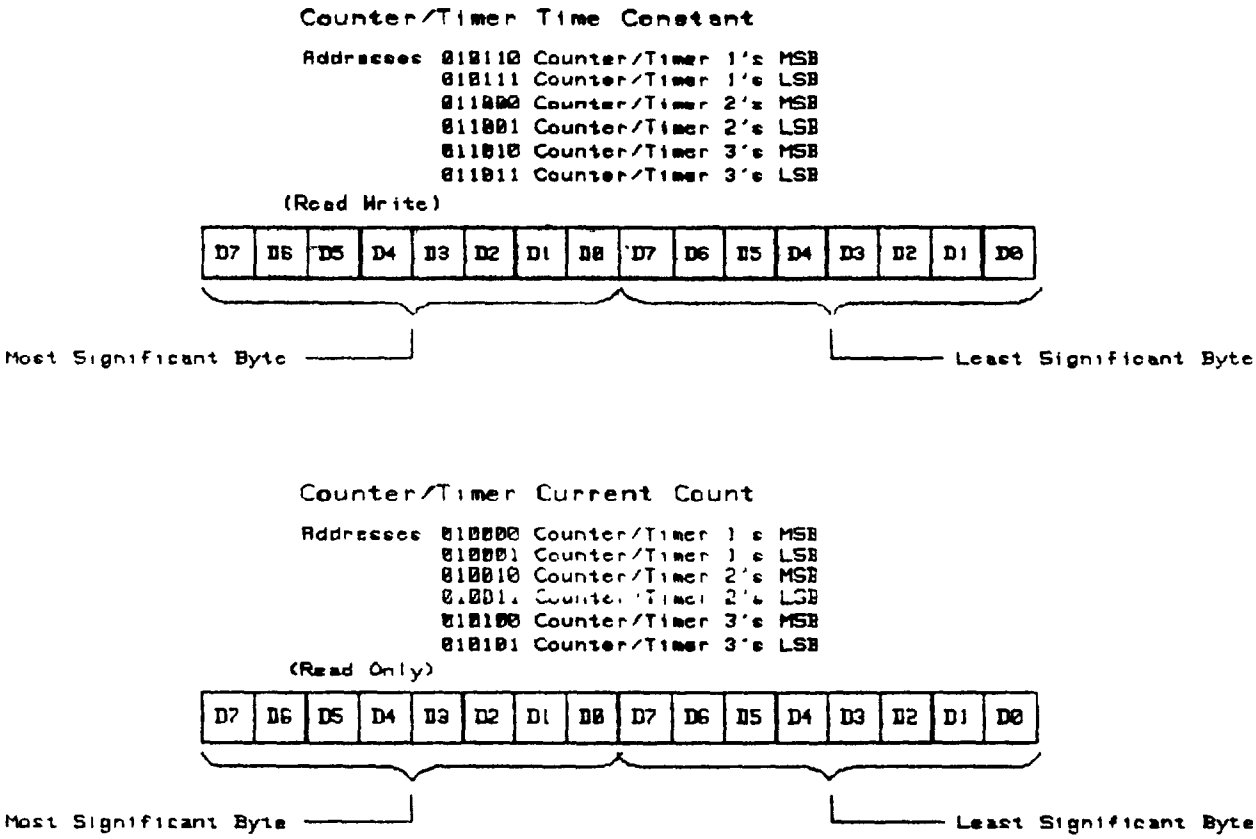


FIGURE 6. Counter/timer registers.

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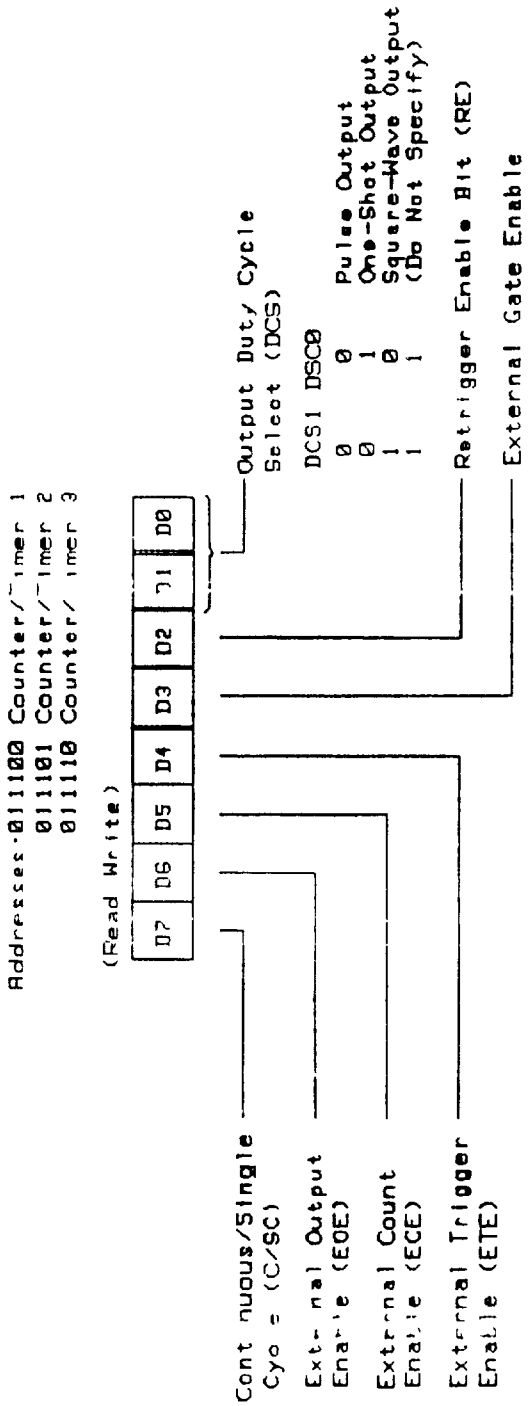
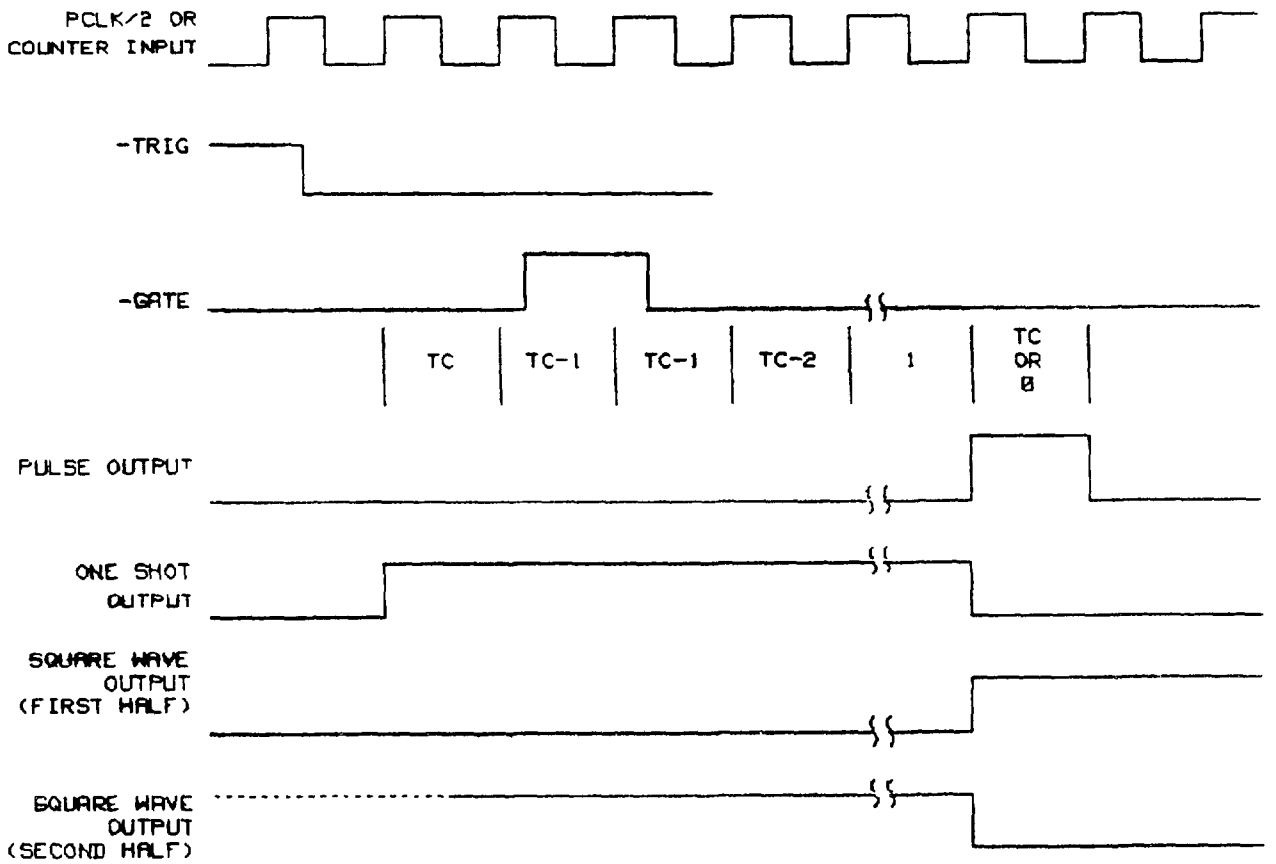


FIGURE 7. Counter/timer mode specification registers.

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TC = Contents of time constant register.

FIGURE 8. Counter/timer waveforms.

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30.12 Port command and status registers. Each of these registers contains the primary command and status bits for its port. Only Port A is available as a general purpose output port in the CEM. The bit assignments for this register is shown on figure 10. A reset forces bit 3 (ORE) to a 1 and other bits to 0. The operation of bits 4-7 are similar to the C/T Command and Status Register bits with the exception that interrupts are associated only with pattern matches in Port A. Bits 2 (IRF) and 3 (ORE) are ignored in the CEM, because the Z8036 ports can only be used as bit ports without handshake. Bit 1 (PMF) is a status bit set to 1 when a pattern match is detected. This bit is not latched and reflects the current state of the pattern match logic. If the port pattern match logic is not enabled (PMS1=PMS0=0 in the Port Mode Specification register), the PMF bit is forced to 0. This is a read only bit; writes to it are ignored. It is defined for Port A only. While bit 0 (IOE) is cleared to a 0, error conditions from pattern recognition logic (a second match before a previous match is acknowledged) are ignored. However, if IOE is 1, such errors will cause IP to be set in the Port Command and Status register and normal operation of the port will be halted until the error condition is corrected.

30.13 Port data register. This register routes the data path from the port to external data inputs/outputs. Only Port A is available as a general purpose output port in the CEM, since Ports B and C are dedicated to counter/timer control functions. The individual bits (figure 11) of the Port A Data register map directly onto the CEM module output pins (i.e., bit 0 to PA0, etc.). A reset does not affect the contents of the register.

30.14 Interrupt vector registers. Each of the Interrupt Vector Registers holds the interrupt vector returned when the source of interrupt is acknowledged. This register is loaded by the user when the CIO is initialized. (See figure 12) The Interrupt Vector register is a read/write register. When read, the value returned always includes the status if MIE=1 (whether or not the associated Vector Includes Status (VIS) bit is 1). If MIE=0, the unmodified vector is returned independent of the state of the VIS bit. A reset does not affect the Interrupt Vector register. The Port B Interrupt Vector register is not available in the CEM, because the B port is dedicated to counter/timer usage. The error status in the Counter/Timer Interrupt Vector register status field indicates that the highest priority counter/timer with an interrupt pending also has its ERR flag set. The microprocessor must poll the Command and Status registers to determine which counter/timer has its ERR flag set.

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30.15 Currrent vector register. When the current vector register is read, it returns the interrupt vector that would have been output by the device during an interrupt acknowledge cycle if its IEI input had been a 1. (See figure 12) The vector returned corresponds to the highest priority IP independent of the IUS. The order of priority (highest to lowest) is: C/T3, Port A, C/T2, Port B (not used in CEM), C/T1. If no enabled interrupts are pending, a pattern of all 1's is output. This register is a read only register. Since a reset disables all interrupts, reading the current vector register after a reset will return all 1's.

30.16 Pattern definition registers. These registers collectively specify the match pattern for Port A only. The pattern specification for each bit in the registers is described on figure 13. Port A can be programmed to generate an interrupt when a specific pattern is recognized at the port. The pattern can be independently specified for each bit as 1,0, rising edge, falling edge or any transition. Individual bits may be masked off. A pattern match is defined as the simultaneous satisfaction of all non-masked bit specifications in the AND mode or the satisfaction of any non-masked bit specifications in either of the OR or OR-Priority Encoded Vector modes. The pattern specified in the Pattern Definition Register assumes that the data path is programmed to be noninverting. Output bits used in the pattern-match logic are internally sampled before the invert/noninvert logic. A reset forces all of these registers to 0. All are read/write.

30.17 Unused Z8036 registers. The following Z8036 registers are unused in the CEM design: Port Handshake Specification Registers (see figure 14). These registers are not available in the CEM, because the A and B ports are defined as bit ports only. The bits in these registers are ignored when the port is a bit port. A reset forces all bits to 0.

30.18 Output crosstalk noise. Output crosstalk noise measurements are taken with the module inputs conditioned so that the output under test is at a stable logic level (1 or 0) and as many other outputs as possible are switching. The maximum voltage limit (AC coupled) for Schottky TTL outputs is 500mV and 400mV for Low Power Schottky TTL outputs. When connected in the worst case mode of operation (i.e., the maximum number of outputs switching simultaneously), the crosstalk noise on all outputs is below the maximum allowable limit. Therefore, no crosstalk user restrictions are defined for this module.

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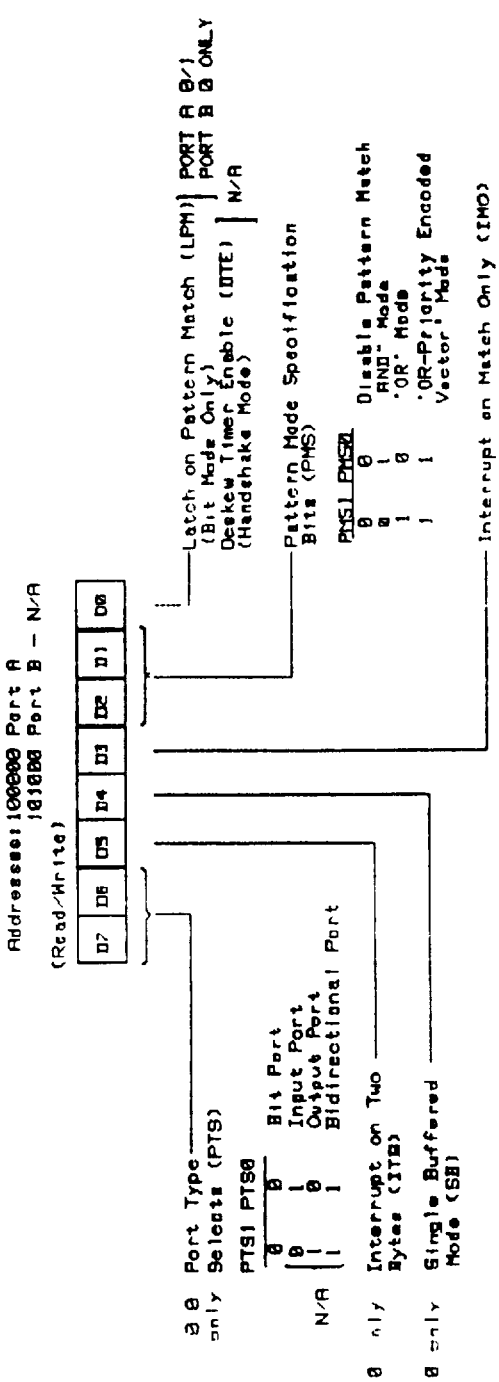
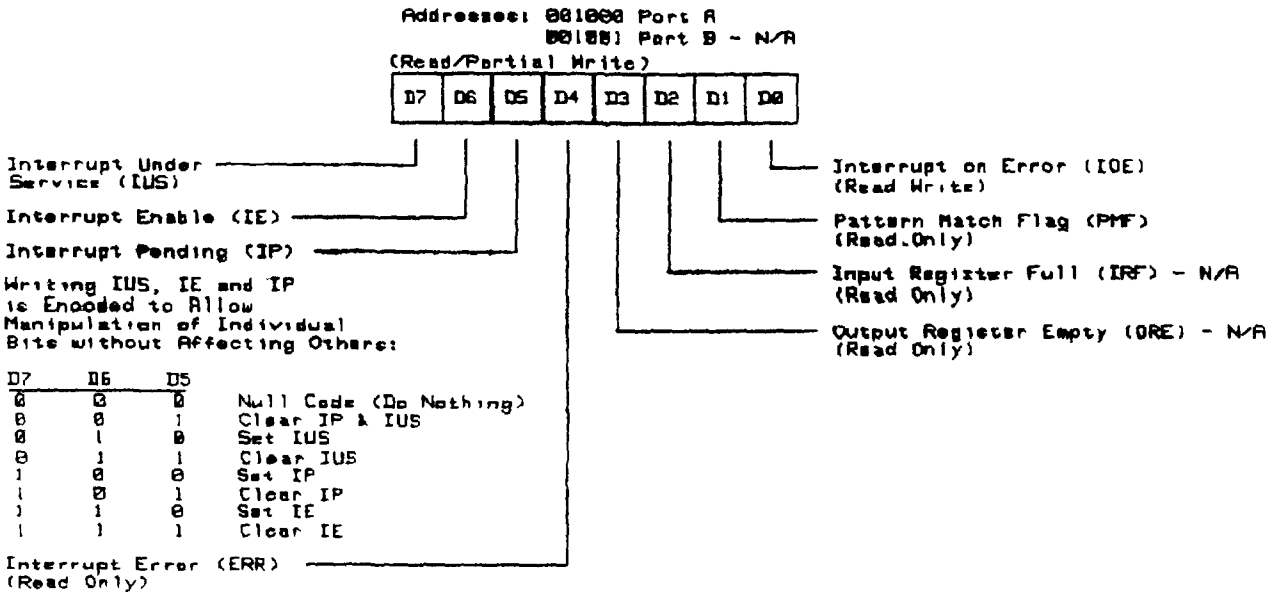


FIGURE 9. Port mode specification registers.

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N/A = not available in CEM design.

FIGURE 10. Port command status registers.

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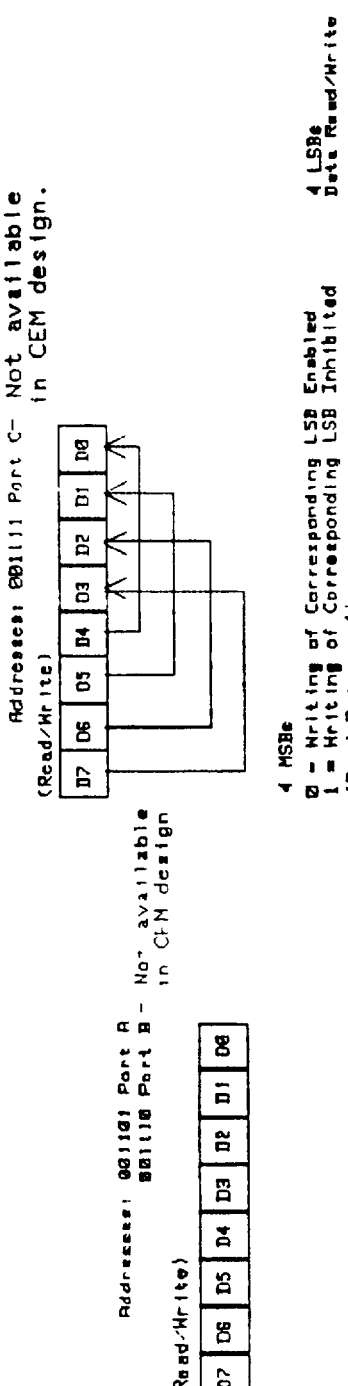


FIGURE 11. Port data registers.

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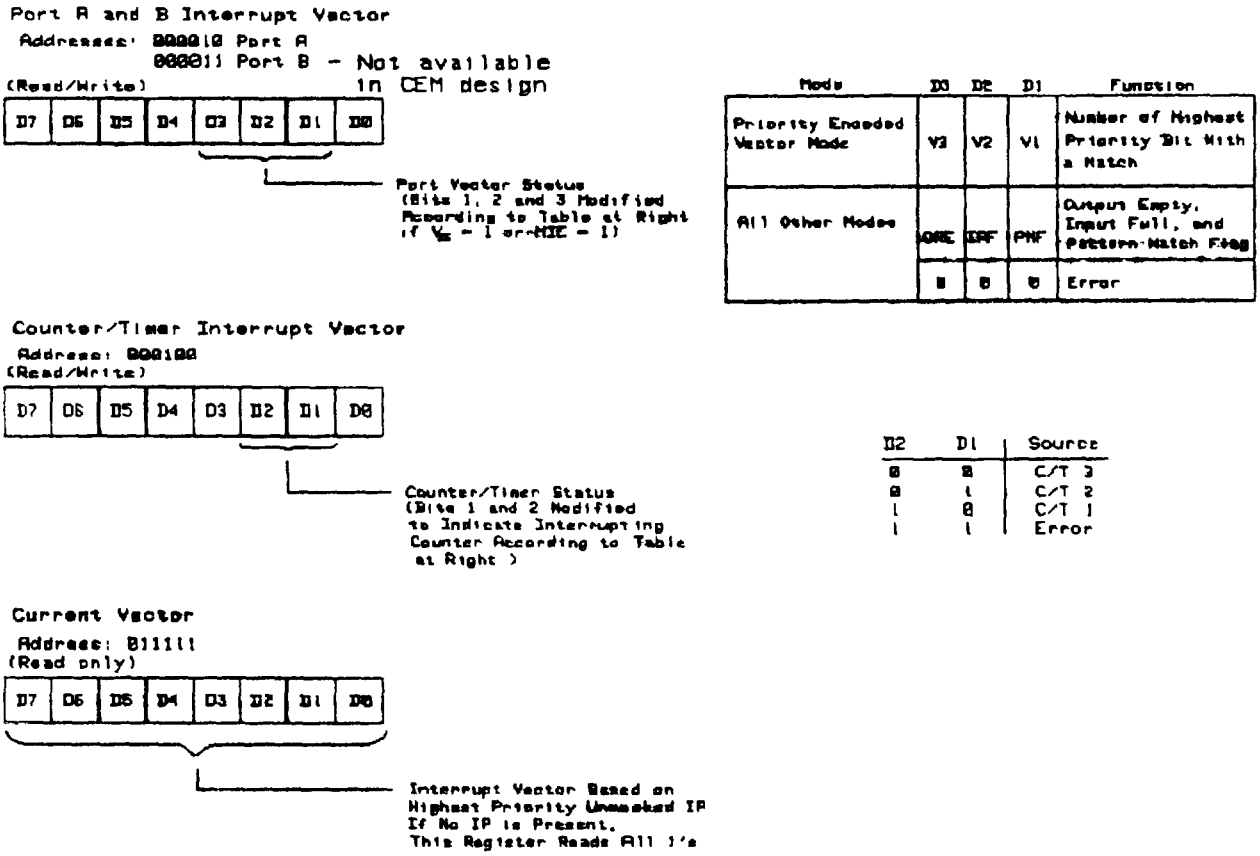
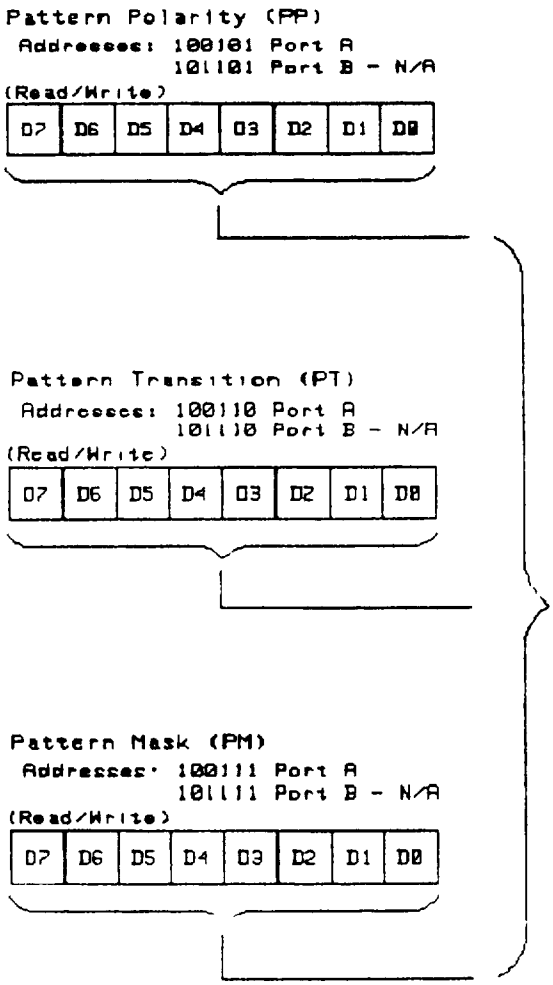


FIGURE 12. Interrupt vector registers.

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PM	PT	PP	Pattern Specification
0	0	X	Bit Masked Off
0	1	X	Any Transition
1	0	0	Zero*
1	0	1	One*
1	1	0	One to Zero Logic Transition*
1	1	1	Zero to One Logic Transition*

*The definition of zero and one is determined by the data path polarity register.

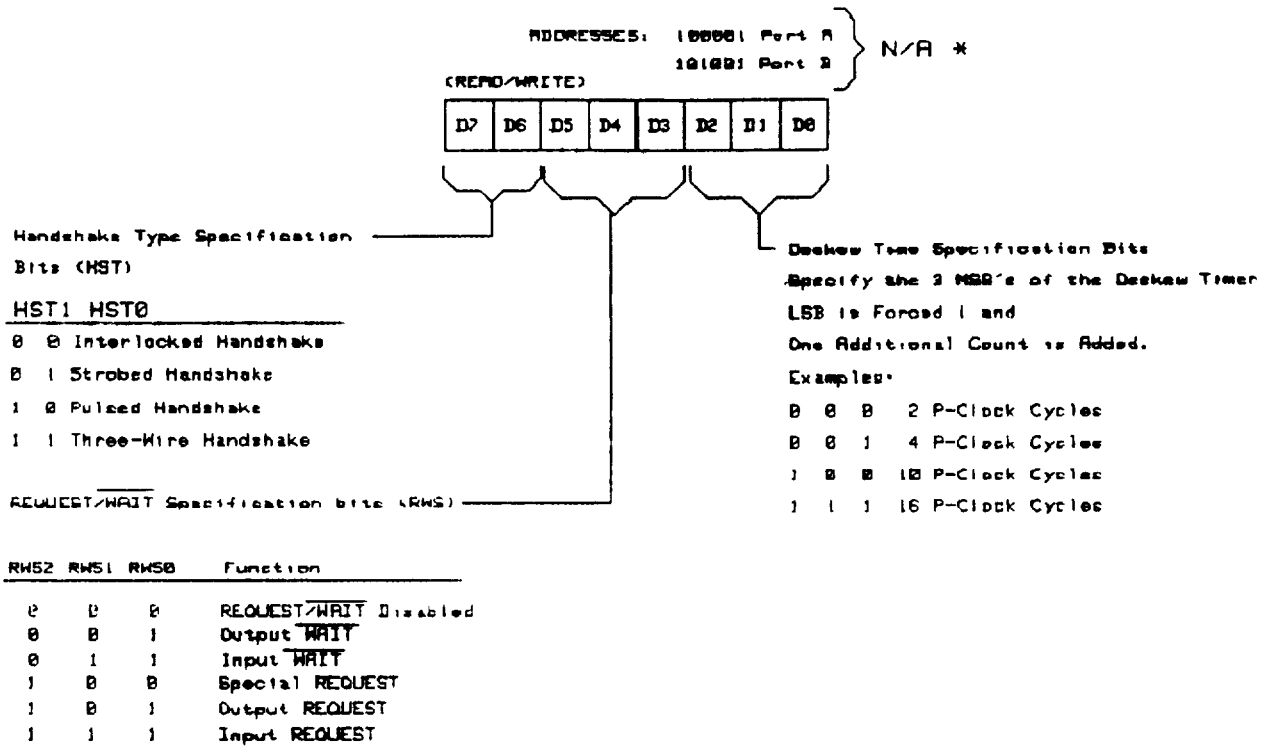
DPP	"ZERO"	"ONE"
0	LOW	HIGH
1	HIGH	LOW

N/A = Not available in CEM design.

FIGURE 13. Pattern definition registers.

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RWS2	RWS1	RWS0	Function
0	0	0	REQUEST/WAIT Disabled
0	0	1	Output WAIT
0	1	1	Input WAIT
1	0	0	Special REQUEST
1	0	1	Output REQUEST
1	1	1	Input REQUEST

N/A = Not available in CEM design
* = Included for reference only.

FIGURE 14. Port handshake specification registers.

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